Experiment 1: Estimator Comparison Against a Non-Visual Baseline

Raw write-up for Chapter 4, Section 4.1

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Reporting cutoff: 6 November 2026

Purpose of this document. This is the raw, complete write-up of Experiment 1, written before the thesis chapter; Section 4.1 of the final report is extracted from it. Experiment 1 was widened on 11 November 2026 to carry the whole estimator comparison: the non-visual estimators it always held, and the visual pipelines that were previously reported separately. It carries both result slots, every table and figure, and the analysis. Every number traces to a file under `output/` named in the slot's "Reporting rules" paragraph. Open concerns are set in red where they arise.

4.1.1 Simulation

Status: completed. Four `dataset_allsensor_realistic` recordings, two layouts each recorded without and with warehouse traffic. Two families of estimator are scored over the full common interval of each: the non-visual ones — wheel odometry, IMU dead reckoning, an EKF at the visual systems' own inertial noise configuration and a measured-noise EKF as diagnostic — and the visual ones, VINS-Fusion once per recording and ORB-SLAM3 with a CPU or GPU feature extractor and its loop closer active or disabled. Traffic is noted as a condition, not tested as a factor. The latency and resource measurement is taken from eighteen repeated runs on the two mapping recordings, three repeats of each of six configurations, and those repeats are the only within-configuration spread the campaign measures.

Objective. Establish how well each estimator localises the robot on the realistic warehouse, whether a non-visual estimator is competitive with the visual pipelines on the same recordings, which of a visual pipeline's own switches decides its error, and at what compute cost. A secondary question is whether each estimator's reported uncertainty describes its error.

Process. Two groups of runs share these recordings.

Non-visual. `script/proprio_estimator.py calibrate` measured each recording's wheel and inertial biases and wheel noise against ground truth from its stationary head and its driven length; `run` produced the wheel, IMU and EKF trajectories offline; `script/analyse_proprio.py` scored them with the live VINS-Fusion and ORB-SLAM3 runs over the common interval. The EKF reported here does *not* use the measured inertial noise: by the design decision of 6 November 2026 its three inertial noise terms are the values both visual configurations declare (`config/wil_sim/stereo_imu.yaml`: gyroscope density 0.002, accelerometer density 0.02, gyroscope bias walk 1×10^{-4}), converted to per-sample σ at 200 Hz by `script/make_vslam_noise_yaml.py`, so that the proprioceptive and visual families run on the same hand-set inertial model; biases and both wheel σ have no visual counterpart and stay measured. The measured-noise filter is run beside it as a diagnostic row and is not the baseline.

Visual. Every run replayed at rate 1.0 from the bag with the node live on the laptop, the deployment condition; no detector ran. VINS-Fusion ran once per recording (run `cpu`). ORB-SLAM3 ran four times per recording: the CPU or the GPU feature extractor (fork patches 10/10b, runs `*_cpu` and `*_gpu`), each with the loop closer active (`loopclosing_*`) or disabled (`noloopclosing_*`), all on 6 November 2026 with the same node binary and configuration (`config/wil_sim/stereo_imu.yaml`). One run per cell, so no within-condition spread exists.

Datasets. Table 1.1: the two static twins and the two traffic recordings of the realistic worlds, both layouts. Traffic is noted as a condition of a recording, not treated as a factor, since no two recordings share a route.

Table 1.1: Recordings of Experiment 1 (names without the `dataset_allsensor_` prefix), from the bags. Route is the ground-truth path length; samples are ground-truth odometry messages.

Recording	Layout	Traffic	Route [m]	Span [s]	Samples
`realistic_000`	U	none (static twin)	171.3	125.5	6 278
`realistic_001`	I	none (static twin)	133.0	96.9	4 847
`realistic_002`	U	pallet jacks and people	102.7	79.5	3 975
`realistic_003`	I	pallet jacks and people	158.0	117.0	5 852

Which recording is which. `realistic_000` (U layout) and `_001` (I layout) are the static twins; `_002` (U) and `_003` (I) carry ten and twelve moving props, pallet jacks on pick–carry–drop jobs and walking people. Routes are 103 to 171 m over 80 to 126 s (the dataset table of the Experiment 1 write-up); no two recordings share a route.

Reporting rules. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed, the convention of every slot; anchored error is the direct difference from ground truth with no alignment and is defined only for the estimators that start from the ground-truth pose; aligned rotation is the geodesic angle after the same fit, defined for every estimator; ANEES is the average normalised estimation error squared over decorrelated 1 Hz samples, 3.0 for a consistent planar filter. Ground truth is the bag's `/ground_truth/odometry` moved from the model origin to the IMU point every estimator reports (lever arm 0.338 m forward, 0.192 m up). The visual runs are pinned by name: VINS-Fusion `cpu`; ORB-SLAM3 `loopclosing_gpu` and `noloopclosing_gpu`, both with the GPU feature extractor, at replay rate 1.0. The loop-closer state is read from each run's `loop_closing.csv` (patch-7 signature: sub-millisecond rows mean a disabled closer), not from the run name. On `realistic_002` the directory named `loopclosing_gpu` first held a run whose manifest names the `realistic_003` bag (3 535 stereo pairs received against the 2 409 that recording holds); it was scored against the wrong truth for a day, is set aside under `trash/output/misfiled_20261106/`, and the run reported here was made on the right bag on the evening of 6 November (2 382 pairs received). Sources: `output/compare/simulation/exp1-realistic/` (`proprio_metrics.csv`, `proprio_rpe_metrics.csv`, the noise models under `noise/`), run directories under `output/output_{wheel,imu,ekf,ekf_measured}/simulation/`. One run per cell, $n = 1$.

Reporting rules, visual arms. Every run is reported on its own. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed, in three dimensions; RPE at 1 s is translational over pairs that span no jump (a step above 10 m s^{-1} between consecutive poses). The loop-closer state is read from `loop_closing.csv` (sub-millisecond rows mean a disabled closer), not from the run name, and every closure the closer accepted is listed with its time and correction. Ground truth is the bag's `/ground_truth/odometry`, moved from the model origin to the IMU point both estimators report (lever arm 0.338 m forward, 0.192 m up) before alignment; the figures draw that moved path. "Kept" is the fraction of the bag's stereo pairs the node processed, so frames lost before the node's own counter

count as lost. The onset analysis uses the start-anchored planar error against a threshold of three times the estimator's median error over its two static runs (VINS-Fusion 1.07 m, ORB-SLAM3 1.31 m). Sources: `output/compare/simulation/exp-realistic-ab/` (`runs_metrics.csv`, `baseline_table.te`
`output/compare/simulation/exp3-timelines/` (`timelines.csv`, `thresholds.csv`), run directories under `output/output_orb/simulation/` and `output/output_vins/simulation/`. The run that first occupied `realistic_002/loopclosing_gpu` had been launched on the `realistic_003` bag with the wrong output path (its manifest names that bag and world, and it received 3 535 stereo pairs where `realistic_002` holds 2 409); it is set aside under `trash/output/misfiled_20261106/` and the cell was re-run on the right bag on the evening of 6 November.

Non-visual estimators

Results. At the visual systems' inertial noise the EKF holds 0.68–2.80 m aligned ATE and is behind VINS-Fusion on all four recordings; the measured-noise diagnostic holds 0.21–0.58 m. Table 1.2 gives every estimator on every recording, table 1.3 the relative pose error at 1 s, and figure 1.1 the trajectories.

Table 1.2: Estimator comparison on the four realistic recordings. The closer state of each ORB-SLAM3 run is read from its artefact. ANEES is reported for the proprioceptive estimators, which carry a covariance; a consistent filter gives 3.0, and every value here is above it. **Bold** marks the lowest aligned ATE of a recording; the measured-noise filter is a diagnostic row, not the baseline, so it is shown bold where it is lowest but is not the slot's result.

Recording	Estimator	Closer	Anch. RMS [m]	Anch. final [m]	Aligned ATE [m]	Aligned rot. [°]	Yaw RMS [°]	ANEES
`realistic_000`	Wheel odometry	–	5.89	5.66	2.56	9.14	18.05	1810
	IMU dead reckoning	–	36.19	76.01	23.02	177.21	0.84	454
	EKF, VSLAM-config noise	–	2.49	3.42	1.34	5.83	10.62	19
	EKF, measured noise (diag.)	–	0.33	0.35	**0.23**	0.47	0.43	111
	VINS-Fusion	–	–	–	0.29	1.85	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	0.29	12.80	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	0.52	12.84	–	–
`realistic_001`	Wheel odometry	–	1.40	2.90	1.03	6.04	7.27	176
	IMU dead reckoning	–	15.40	39.82	13.05	0.25	0.55	160
	EKF, VSLAM-config noise	–	0.94	0.83	0.68	3.81	4.51	8
	EKF, measured noise (diag.)	–	0.26	0.26	**0.21**	0.33	0.36	31
	VINS-Fusion	–	–	–	0.22	1.18	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	4.78	18.98	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	0.40	15.13	–	–
`realistic_002`	Wheel odometry	–	6.54	6.08	4.81	17.03	25.00	1778
	IMU dead reckoning	–	25.88	39.17	19.01	118.74	1.91	695
	EKF, VSLAM-config noise	–	2.06	3.29	1.58	9.92	13.65	37
	EKF, measured noise (diag.)	–	0.59	0.30	0.37	0.70	1.03	191
	VINS-Fusion	–	–	–	0.66	3.57	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	0.40	16.04	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	**0.17**	16.08	–	–
`realistic_003`	Wheel odometry	–	6.71	7.19	4.30	19.24	35.82	3399
	IMU dead reckoning	–	13.72	35.47	11.60	180.00	1.68	140
	EKF, VSLAM-config noise	–	4.44	6.23	2.80	15.92	28.01	158
	EKF, measured noise (diag.)	–	0.79	1.49	0.58	3.90	5.72	106
	VINS-Fusion	–	–	–	**0.27**	1.51	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	0.50	13.16	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	1.91	13.53	–	–

Analysis. With both families on the same hand-set inertial model the ordering is settled: VINS-Fusion is the most accurate estimator on every recording (0.22–0.66 m) and the EKF, at 0.68–2.80 m, is behind it on all four and behind the better ORB-SLAM3 run on three. The measured-noise diagnostic puts the same filter at 0.21–0.58 m, ahead of VINS-Fusion on three of four, so the whole of the EKF's deficit is the noise model and none of it the architecture — the September control, reproduced on new recordings and larger.

The mechanism is the one isolated in September. The configuration's gyroscope density is five to

Table 1.3: Relative pose error at $\Delta t = 1\,\mathrm{s}$, RMSE over all pairs and over pairs that span no detected jump (a step above $10\,\mathrm{m\,s^{-1}}$ between consecutive poses); the jump count is per run. **Bold** marks the lowest jump-free translational RPE of a recording.

Recording	Estimator	Jumps	Translation [m]		Rotation [°]	
			all pairs	clean	all pairs	clean
`realistic_000`	Wheel odometry	0	0.028	0.028	1.720	1.720
	IMU dead reckoning	0	1.941	1.941	0.023	0.023
	EKF	44	0.240	0.218	0.979	0.893
	EKF	0	0.039	0.039	0.098	0.098
	VINS-Fusion	0	0.021	0.021	0.443	0.443
	ORB-SLAM3 GPU, LC run	3	0.246	0.017	5.902	0.094
	ORB-SLAM3 GPU, no-LC run	3	0.254	**0.013**	5.909	0.090
`realistic_001`	Wheel odometry	0	0.033	0.033	1.733	1.733
	IMU dead reckoning	0	0.980	0.980	0.023	0.023
	EKF	42	0.254	0.202	0.964	0.811
	EKF	0	0.045	0.045	0.090	0.090
	VINS-Fusion	0	0.018	**0.018**	0.366	0.366
	ORB-SLAM3 GPU, LC run	7	1.191	0.034	7.031	0.120
	ORB-SLAM3 GPU, no-LC run	2	0.302	0.020	6.845	0.155
`realistic_002`	Wheel odometry	0	0.027	0.027	1.997	1.997
	IMU dead reckoning	0	1.505	1.505	0.033	0.033
	EKF	67	0.243	0.158	0.985	0.780
	EKF	0	0.035	0.035	0.075	0.075
	VINS-Fusion	0	0.038	0.038	0.790	0.790
	ORB-SLAM3 GPU, LC run	4	0.463	**0.017**	10.403	0.084
	ORB-SLAM3 GPU, no-LC run	1	0.147	0.028	7.422	0.070
`realistic_003`	Wheel odometry	0	0.023	0.023	1.447	1.447
	IMU dead reckoning	0	0.936	0.936	0.023	0.023
	EKF	22	0.202	0.185	1.026	0.969
	EKF	0	0.036	0.036	0.154	0.154
	VINS-Fusion	0	0.014	**0.014**	0.327	0.327
	ORB-SLAM3 GPU, LC run	3	0.165	0.028	6.068	0.174
	ORB-SLAM3 GPU, no-LC run	3	0.151	0.036	6.088	0.333

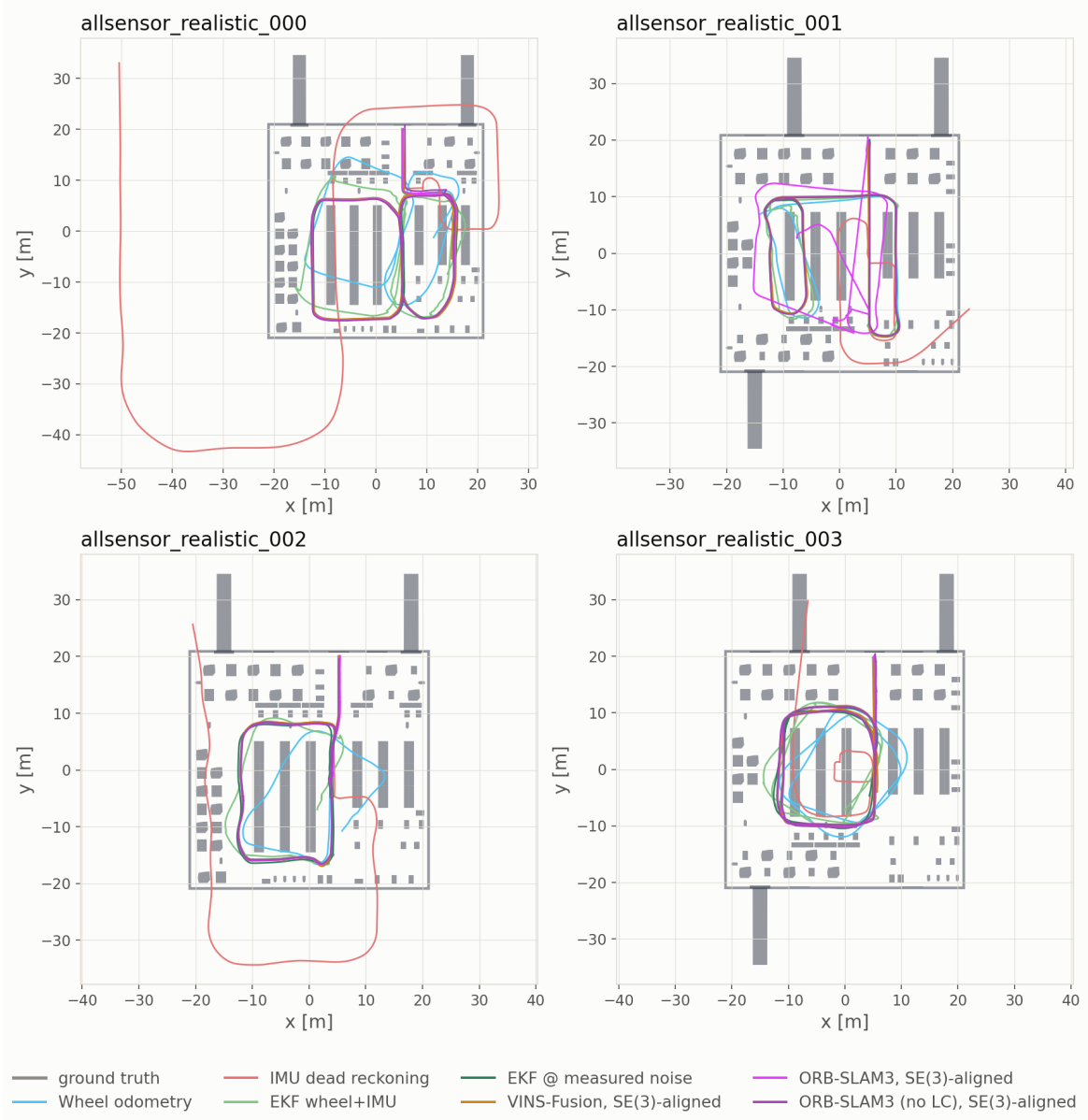


Figure 1.1: Estimator comparison on every realistic recording, world frame: the dead-reckoning estimators as estimated from the ground-truth start pose, the visual estimators after the rigid $SE(3)$ alignment used for their metrics, one legend for all four panels. The IMU leaves the building on every recording; the EKF at the visual systems' noise (light green) wanders in heading where the measured-noise filter (dark green) stays on the route; ORB-SLAM3's divergent runs are the excursions on `001` (loop closer active) and `003` (closer disabled).

seven times the measured value, which inflates the heading process noise forty-fold; with the heading unobserved except through the wheel yaw-rate residual, that residual steers the heading instead of the bias, and yaw RMS rises from 0.4–5.7° to 4.5–28.0°. On routes of 100 to 170 m with repeated turns that heading error is what the aligned ATE measures.

ORB-SLAM3's divergences are not a loop-closure effect. With the GPU front end six of eight runs hold 0.17–0.52 m and two diverge, to 4.78 m with the closer active on `realistic_001` and to 1.91 m with it disabled on `realistic_003`, while the other closer state on each of those recordings holds 0.40 and 0.50 m. Each divergent run carries three or seven jumps (table 1.3) and its clean relative error is as low as the stable runs', so the error is a few discrete events, not steady-state tracking. Which front-end condition produces them is not isolated here; the CPU runs of the same recordings are scored below, where the same pattern recurs.

Traffic appears as a condition, not a tested factor. VINS-Fusion is 0.66 m on the U traffic recording against 0.29 m on its static twin and 0.27 m against 0.22 m on the I layout; the proprioceptive estimators are unaffected by construction. The U recordings do not share a route, so the difference cannot be credited to traffic alone.

The complementary failure of the two proprioceptive sensors holds. The IMU keeps heading to 0.5–1.9° RMS and loses position by 12–23 m; the wheels keep position to 1.0–4.8 m and lose 7–36° of heading. Every proprioceptive covariance is overconfident: ANEES 8–3399 against 3.0, the EKF at the configuration noise least so because its inflated process noise is closer to the error it actually makes.

Visual estimators

Results. VINS-Fusion holds 0.22–0.66 m aligned ATE on the four recordings with no jump and 1.2–3.6° of rotation error; its worst recording is the U layout with traffic, 0.66 m against 0.29 m on that layout's static twin, while on the I layout traffic changes nothing (0.27 against 0.22 m). ORB-SLAM3 spans 0.14–6.02 m: eleven of its sixteen runs lie between 0.14 and 0.52 m and five diverge, and which five is decided by the loop closer and the recording, not by the front end (table 1.4, figures 1.2 and 1.3). Neither front end loses tracking on any recording, with or without traffic: zero tracking-loss events and zero resets in twenty runs.

Two columns of table 1.4 qualify those RMSE figures. The first is the worst single error each run reaches after its initialisation transient, which the RMSE averages away: over the fifteen runs that held it is 0.26–1.25 m, between 0.8 and 2.6 times the run's own RMSE, so no run that tracked was ever more than about 1.25 m from the truth; over the five that diverged it is 2.25–13.42 m. The two orderings are not the same one. On `realistic_000` VINS-Fusion has the lower RMSE (0.290 against 0.338 m) and ORB-SLAM3 the smaller worst case (0.260 against 0.576 m), so which estimator is "better" on that recording depends on whether the question is average accuracy or the largest excursion — and for a vehicle in a 3.0 m aisle it is the second. The exclusion matters: every ORB-SLAM3 run peaks near 2.5 s at three to four metres on the gravity re-alignment that ends its inertial initialisation, and a raw maximum would report that initialisation as though it were driving.

The second column is the same ATE as a percentage of the ground-truth path length. It exists so that these 103–172 m routes can be set beside the real robot's much shorter ones, and it separates the two outcomes more sharply than the metre figure does: 0.09–0.64 % for the fifteen runs that held against 1.11–4.51 % for the five that diverged. It is a ratio of two quantities already in the table, not a KITTI-style sub-sequence drift rate, and it is not comparable to one.

Analysis: the loop closer decides, in both directions. On the static I layout both loop-closing runs accept one closure at 73.2 s, in the heading change at 76 % of the route, with a correction of 14.7 m (CPU) and 13.3 m (GPU) and 1.9–2.8°; a global bundle adjustment follows and the trajectory is 6.02 and 4.78 m off for the rest of the run, where the two closure-free runs on the same recording hold 0.50 and 0.40 m

Table 1.4: Visual estimators on the four realistic-warehouse recordings, 6 November 2026, replay rate 1.0. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed; RPE is translational, at $\Delta = 1$ s over pairs that do not span a jump; "Jumps" counts steps above 10 m s^{-1}; "Kept" is the fraction of the bag's stereo pairs the estimator processed (frames lost before the node's own queue count as lost). Loop-closer state is read from `loop_closing.csv`, not from the run name. Static and dynamic rows of one layout share a world geometry and differ in traffic only. "Max" is the largest single position error the run reaches *after* its initialisation transient — the same cut the onset analysis of table 1.7 makes, after the last jump inside the first 15 s — on the same alignment as the ATE column, and records the excursion that the RMSE averages away. Every ORB-SLAM3 run peaks near 2.5 s on the gravity re-alignment that ends its inertial initialisation, at three to four metres; that peak is excluded here because it measures the initialisation and not the driving. "ATE/path" is the aligned ATE as a percentage of the ground-truth path length, so that runs over routes of different length can be compared; it is a ratio, not a sub-sequence drift rate. **Bold** marks the lowest ATE, the lowest Max and the lowest jump-free RPE among the five arms of a recording; with one run per arm it records which run came lowest, not that the arm is better.

Recording	Layout	Scene	Estimator	ATE [m]	Max [m]	ATE/path [%]	Rot. [°]	$RPE_{1\,s}$ [m]	Jumps	Kept	Resets
`realistic_000`	U	static	ORB-SLAM3	0.338	**0.260**	0.20	13.74	0.083	2	0.72	0
			ORB-SLAM3 GPU	0.293	0.525	0.17	12.80	0.079	3	0.99	0
			ORB-SLAM3 (no LC)	0.483	1.249	0.28	14.05	0.090	2	0.74	0
			ORB-SLAM3 GPU (no LC)	0.523	0.995	0.30	12.84	**0.078**	3	0.99	0
			VINS-Fusion	**0.290**	0.576	0.17	1.85	0.084	0	0.77	0
`realistic_001`	I	static	ORB-SLAM3	6.021	12.591	4.51	16.92	0.094	6	0.73	0
			ORB-SLAM3 GPU	4.780	13.421	3.58	18.98	0.092	7	0.99	0
			ORB-SLAM3 (no LC)	0.496	0.642	0.37	16.05	0.093	1	0.74	0
			ORB-SLAM3 GPU (no LC)	0.400	0.514	0.30	15.13	**0.088**	2	0.97	0
			VINS-Fusion	**0.221**	**0.456**	0.17	1.18	0.089	0	0.78	0
`realistic_002`	U	dynamic	ORB-SLAM3	1.393	2.254	1.35	16.99	**0.078**	4	0.77	0
			ORB-SLAM3 GPU	0.400	0.595	0.39	16.04	**0.078**	4	0.98	0
			ORB-SLAM3 (no LC)	0.293	0.501	0.28	17.12	0.080	2	0.83	0
			ORB-SLAM3 GPU (no LC)	**0.170**	**0.255**	0.16	16.08	0.083	1	0.99	0
			VINS-Fusion	0.663	1.136	0.64	3.57	0.116	0	0.88	0
`realistic_003`	I	dynamic	ORB-SLAM3	**0.139**	**0.316**	0.09	13.52	0.075	2	0.94	0
			ORB-SLAM3 GPU	0.504	0.756	0.32	13.16	0.076	3	0.99	0
			ORB-SLAM3 (no LC)	1.755	2.741	1.11	14.22	0.080	2	0.88	0
			ORB-SLAM3 GPU (no LC)	1.911	2.800	1.21	13.53	0.083	3	0.99	0
			VINS-Fusion	0.268	0.537	0.17	1.51	**0.071**	0	0.87	0

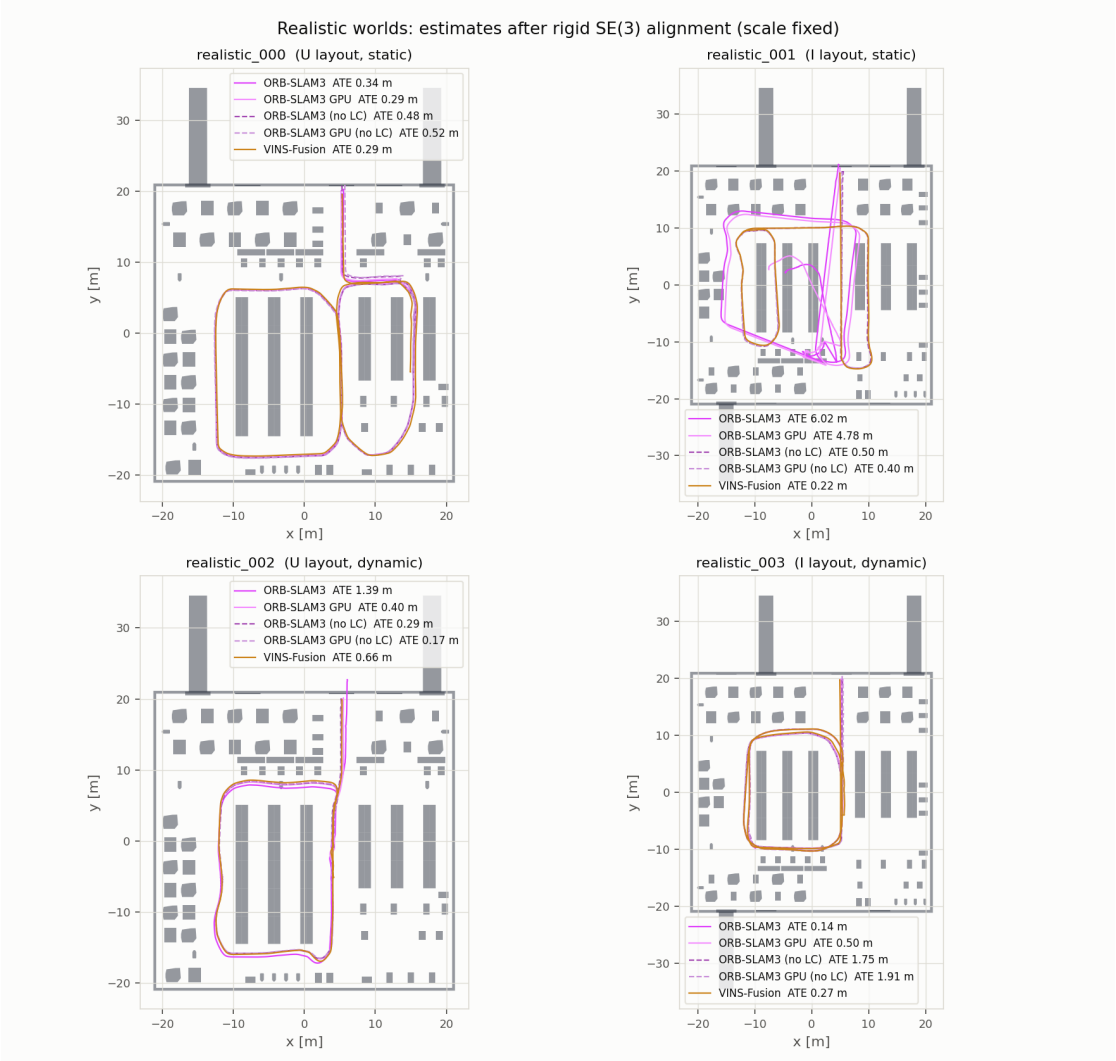


Figure 1.2: Unfiltered estimates on the four realistic-warehouse recordings after rigid $SE(3)$ alignment with scale fixed, on the warehouse footprint. Top: the static twins; bottom: the same layouts with traffic. Solid magenta is ORB-SLAM3 with its loop closer active, dashed with it disabled, lighter for the GPU front end; orange is VINS-Fusion. The two excursions on `realistic_001` are the loop-closing runs after their false closure; the 1.4 m run on `realistic_002` is the CPU loop-closing run; the closure-free runs on `realistic_003` lie on the route in plan and diverge vertically.

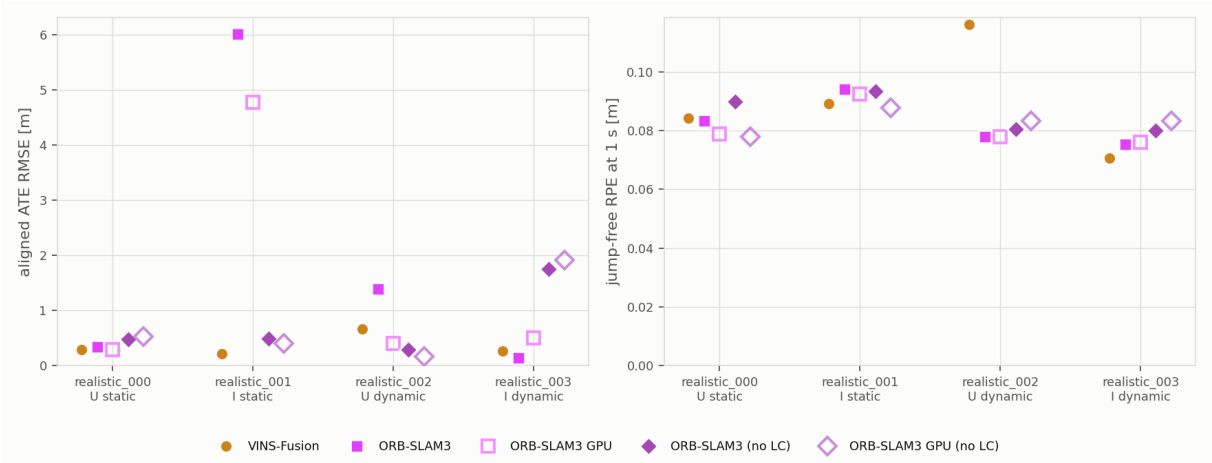


Figure 1.3: Aligned ATE (left) and jump-free translational RPE at 1 s (right) of every visual run, one marker per run; filled markers are the CPU front end, open markers the GPU front end of the same configuration. The relative error is 0.07–0.12 m for every run, so the ATE differences are discrete events, not tracking quality.

(figure 1.5). A route with half a metre of accumulated error cannot need a fourteen-metre correction, so the match is false: the I layout's long parallel aisles present the same rack faces twice, and the matched keyframes (97 against 255, 104 against 272) are one aisle apart. The same candidate was then re-proposed four times and rejected by the inertial geometry check, after the damage. That both front ends accept it at the same frame with the same correction says the place-recognition stage, not the extractor, is where the fault lies.

On the I layout with traffic the sign reverses. Both loop-closing runs accept one closure at 56.5 s (48 % of the route) with a 0.33 m correction and hold 0.14 and 0.50 m; the closure-free runs read 1.76 and 1.91 m, yet their planar error never exceeds 0.7 m (figure 1.7). The difference is vertical: the closure-free estimates climb to 4.7 and 5.1 m above the floor by the end of the run while the loop-closing runs stay within 0.6 and 1.4 m, and the three-dimensional ATE reports that climb. The GPU loop-closing run on this recording also rejected 148 further candidates by inertial geometry, every one of them after its single accepted closure. On the U layout the closer is nearly inert: two accepted closures per run on `realistic_000` with corrections of 0.07–0.12 m, one of 0.26–0.33 m per run on `realistic_002`, and the eight U-layout ORB-SLAM3 runs within 0.17–0.52 m except the CPU loop-closing run on `realistic_002`, whose 1.39 m is an offset present from 6.5 s on, before any closure (figure 1.6); with four jumps and the lowest relative error on the recording it is an initialisation event, not a tracking one, and the GPU run of the same configuration, which accepts the same closure at 65 s, does not reproduce it (0.40 m).

Analysis: the front end changes throughput, not accuracy. The GPU extractor processes 97–99 % of each bag's frames where the CPU extractor processes 72–94 %, and its tracking median is 30–37 ms against 49–55 ms (from the per-run `performance.csv`). Across the eight CPU/GPU pairs with the same closer state the ATE differs by at most 0.16 m in five; the three larger differences (−1.2, −1.0 and 0.4 m) are each one run's discrete event, in both directions, and the two divergence patterns above occur in both front ends alike. At this frame budget the extractor is not the limiting stage of accuracy on these recordings.

Analysis: traffic as a condition. VINS-Fusion is 0.37 m worse on the U layout with traffic than on its static twin and 0.05 m worse on the I layout; its relative error on `realistic_002` is the highest of any run (0.116 m) and its error trace carries two waves of 0.7 m at 25 and 45 s on the western and southern aisles, where the pallet jacks work, that no static run shows (figure 1.6). ORB-SLAM3's closure-free runs are *better* on the U traffic recording (0.29, 0.17 m) than on its static twin (0.48, 0.52 m). With one run per cell and routes that differ between twin and traffic recording, neither number is a traffic effect; what the

slot establishes is that physically simulated traffic degrades the tracker that follows features longest, and breaks neither.

Analysis: the two estimators err in perpendicular directions. VINS-Fusion drifts sideways and ORB-SLAM3 drifts forwards, which the aligned ATE cannot show. Decomposing each run's error in the ground-truth heading frame (, figure 1.8), VINS-Fusion's cross-track share of the planar error is 0.67–0.73

Table 1.5: Translation error of the visual arms split in the ground-truth heading frame, on the same aligned series the reported ATE is computed from, so that the parts account for the whole. "Along" is the component along the truth's heading and "cross" the one perpendicular to it; "bias" is the mean of the signed component, so a bias near the RMS means a constant offset rather than scatter. "Cross share" is the cross-track fraction of the planar squared error: 0.5 is isotropic and above it the estimate drifts sideways, which is the direction that matters in a 3.0 m aisle. One run per cell; ORB-SLAM3's run-to-run spread reaches 0.277 m, so differences smaller than that between single runs are not evidence.

Recording	Estimator	ATE [m]	Along [m]	Cross [m]	Vert. [m]	Along bias	Cross bias	Cross share
`realistic_000`	VINS-Fusion	0.290	0.148	0.211	0.133	-0.028	+0.074	0.67
	ORB-SLAM3	0.338	0.281	0.095	0.162	-0.072	+0.029	0.10
	ORB-SLAM3 GPU	0.293	0.237	0.093	0.145	-0.077	+0.019	0.13
	ORB-SLAM3 (no LC)	0.483	0.412	0.185	0.171	-0.098	+0.035	0.17
	ORB-SLAM3 GPU (no LC)	0.523	0.422	0.263	0.161	-0.073	+0.063	0.28
`realistic_001`	VINS-Fusion	0.221	0.070	0.107	0.180	-0.011	-0.044	0.70
	ORB-SLAM3	6.021	3.687	4.756	0.213	-1.598	-1.372	0.62
	ORB-SLAM3 GPU	4.780	2.644	3.978	0.182	-0.909	-1.166	0.69
	ORB-SLAM3 (no LC)	0.496	0.259	0.093	0.413	-0.079	+0.036	0.11
	ORB-SLAM3 GPU (no LC)	0.400	0.196	0.046	0.346	-0.051	+0.022	0.05
`realistic_002`	VINS-Fusion	0.663	0.236	0.091	0.613	-0.019	+0.030	0.13
	ORB-SLAM3	1.393	1.311	0.454	0.118	-0.297	+0.044	0.11
	ORB-SLAM3 GPU	0.400	0.328	0.148	0.175	-0.081	-0.008	0.17
	ORB-SLAM3 (no LC)	0.293	0.222	0.100	0.163	+0.001	-0.048	0.17
	ORB-SLAM3 GPU (no LC)	0.170	0.100	0.051	0.127	-0.002	-0.032	0.21
`realistic_003`	VINS-Fusion	0.268	0.101	0.167	0.184	+0.035	+0.020	0.73
	ORB-SLAM3	0.139	0.070	0.042	0.112	-0.011	-0.012	0.26
	ORB-SLAM3 GPU	0.504	0.104	0.045	0.491	-0.003	-0.022	0.16
	ORB-SLAM3 (no LC)	1.755	0.072	0.071	1.752	-0.014	-0.005	0.50
	ORB-SLAM3 GPU (no LC)	1.911	0.199	0.114	1.897	-0.042	+0.016	0.25

on three of the four recordings while every ORB-SLAM3 run that held tracking lies between 0.11 and 0.25. On `realistic_000` the two score 0.293 and 0.290 m: indistinguishable, and perpendicular. For a 0.83 by 0.50 m robot in a 3.0 m aisle they are not equally safe, because cross-track error closes the gap to the racking and along-track error does not.

The false closure is also the failure that throws the estimate sideways. The two loop-closing runs on `realistic_001` are the only ORB-SLAM3 runs whose cross-track share exceeds 0.5 (0.62 and 0.69), with cross-track RMS of 4.76 and 3.98 m; of the five diverged runs only those two are cross-track dominated, so the direction belongs to the false closure and not to divergence in general. The error is scatter rather than offset throughout: the mean of each signed component is 0.17–0.36 of its RMS in every arm. Both decompositions are computed on the same aligned series the reported ATE uses, so the parts account for the whole, and both are pooled over the twenty runs rather than ranked, because one run per cell cannot separate arms that differ by less than ORB-SLAM3's 0.277 m run-to-run spread.

Analysis: the rotation column. ORB-SLAM3's 13–19° rotational RMSE is one event on every run: the first jump lies at 2.5–4.8 s (table 1.7), where inertial initialisation re-aligns the world frame to gravity, and after it the clean rotational RPE is 0.07–0.33°. The jump detector is translational, so the column includes the event; the median is the number to read. Splitting the error by axis confirms what kind of event it is (, figure 1.9): yaw, pitch and roll RMS are all 14–19° while their medians are 0.88–1.51°, and an error that moves all three axes together and persists in none is a frame re-alignment rather than drift. Turning does not raise the rotation error either: pooled over the twenty runs the median while turning exceeds the

Table 1.6: Orientation error of the visual arms. Yaw, pitch and roll are the wrapped per-axis differences between the aligned estimate and the truth; for a robot on a flat floor a sustained pitch or roll error is a gravity-alignment defect rather than drift. The median is given beside the RMS because the RMS of an ORB-SLAM3 run is dominated by the single re-alignment at its inertial initialisation. "Turning" is a ground-truth turn rate above $10\,^\circ\,s^{-1}$, the threshold the onset analysis uses; the two medians are paired within a run. One run per cell.

Recording	Estimator	Yaw [°]	Pitch [°]	Roll [°]	Rot. RMS	Rot. med.	Med. turning	Med. straight
`realistic_000`	VINS-Fusion	0.45	1.20	1.34	1.85	0.85	0.80	0.89
	ORB-SLAM3	23.60	13.66	23.58	13.74	0.28	0.30	0.28
	ORB-SLAM3 GPU	18.75	12.75	18.71	12.80	0.33	0.37	0.32
	ORB-SLAM3 (no LC)	19.99	14.01	19.98	14.05	0.25	0.28	0.25
	ORB-SLAM3 GPU (no LC)	23.26	12.74	23.27	12.84	0.49	0.48	0.49
`realistic_001`	VINS-Fusion	0.24	0.90	0.72	1.18	0.93	0.86	0.96
	ORB-SLAM3	24.27	15.16	24.39	16.92	6.83	6.82	6.83
	ORB-SLAM3 GPU	26.16	14.55	24.24	18.98	6.57	6.58	6.56
	ORB-SLAM3 (no LC)	17.02	15.05	17.31	16.05	3.52	3.44	3.52
	ORB-SLAM3 GPU (no LC)	17.08	14.43	17.19	15.13	2.62	2.61	2.62
`realistic_002`	VINS-Fusion	0.21	1.65	3.15	3.57	2.74	2.55	2.77
	ORB-SLAM3	2.06	16.95	1.93	16.99	1.31	1.33	1.30
	ORB-SLAM3 GPU	14.51	15.94	14.46	16.04	0.59	0.62	0.56
	ORB-SLAM3 (no LC)	19.80	17.02	19.75	17.12	0.62	0.65	0.60
	ORB-SLAM3 GPU (no LC)	12.13	16.05	12.09	16.08	0.34	0.44	0.32
`realistic_003`	VINS-Fusion	0.24	0.70	1.31	1.51	1.04	0.99	1.06
	ORB-SLAM3	14.97	13.44	14.91	13.52	0.45	0.47	0.44
	ORB-SLAM3 GPU	10.39	12.99	10.49	13.16	1.39	1.61	1.24
	ORB-SLAM3 (no LC)	15.56	13.70	15.70	14.22	2.24	2.07	2.24
	ORB-SLAM3 GPU (no LC)	13.54	13.04	13.68	13.53	2.54	2.60	2.52

median while straight by 0.01°, and eleven of twenty runs are positive, which is a coin flip. VINS-Fusion carries a smaller signature of the same kind, pitch and roll RMS of 1.05 and 1.33° against a yaw RMS of 0.24°; whether that is a residual gravity error or the rigid fit absorbing a tilt has not been separated, and it is reported as an observation.

Latency and resources. By the design decision of 6 November 2026 the real-time and resource measurement of this experiment is taken only on the two mapping recordings, each configuration run three times: `dataset_allsensor_realistic_map` (U layout, 286 s, 8 725 stereo pairs) and `dataset_allsensor_realistic_01_map` (I layout, 315 s, 9 552 pairs). Eighteen runs exist, six configurations of three repeats each at replay rate 1.0 with no detector: ORB-SLAM3 with the CPU and with the GPU front end, loop closer disabled, and VINS-Fusion (table 1.8). The repeats are tight, front-end medians within 1–7 ms of their mean and peak memory within 30 Mi B.

The two pipelines divide the cost differently, and neither is cheaper than the other outright. VINS-Fusion optimises in 44 ms where ORB-SLAM3's local mapping takes 211–406 ms, and holds its state in a quarter of the memory, but spends more than twice the processor time: about eight cores of one against three and a half. Their front ends are within 6 ms of each other, and every configuration keeps 75–96 % of the bag's frames, so the choice between them is a memory-against-cores trade, not a speed one.

Between ORB-SLAM3's two front ends the recordings disagree. On the U recording the GPU extractor behaves as the single runs showed, tracking in 37 against 42 ms and keeping 96 against 93 % of frames; on the I recording the sign reverses, 52 against 44 ms and 75 against 91 % kept, with the local-mapping median at 406 ms. VINS-Fusion reads the same on both recordings (front end 38.1 and 38.6, back end 44.3 and 44.5 ms), so the recordings themselves do not impose a difference of that size.

[Confound: run order] The six GPU runs were made consecutively from 21:04 to 22:04 and the six CPU runs from 23:27 to 00:18, so the I-recording GPU repeats were the fourth to sixth GPU runs of a session on a laptop that throttles at 90 °C; the front-end record notes that timing comparisons need a cool-start, interleaved order. The six VINS-Fusion runs are a separate session of 7 November and are not a control for that state. The U-recording figures are the ones to quote until the I recording is repeated interleaved.

Table 1.7: Realistic world, unfiltered runs: onset of sustained error and its place on the route, per run. Onset is the first time the start-anchored position error stays above 3× the estimator's median error over its static runs (VINS-Fusion 1.07 m, ORB-SLAM3 1.31 m) for the rest of the run; "where" is the route segment and the share of the route completed at that time; the first jump is a step above 10 m s^{-1} between consecutive poses. "never" means the error stayed below the threshold for the whole run.

Recording	Estimator	LC	Onset [s] (where)	First jump [s]
`realistic_000`	ORB-SLAM3	active	never	4
	ORB-SLAM3	disabled	never	4
	ORB-SLAM3 GPU	active	never	4
	ORB-SLAM3 GPU	disabled	60 (straight 48 %)	4
	VINS-Fusion	—	never	—
`realistic_001`	ORB-SLAM3	active	73 (turn 78 %)	5
	ORB-SLAM3	disabled	never	5
	ORB-SLAM3 GPU	active	73 (turn 78 %)	5
	ORB-SLAM3 GPU	disabled	never	5
	VINS-Fusion	—	never	—
`realistic_002`	ORB-SLAM3	active	6 (straight 6 %)	3
	ORB-SLAM3	disabled	never	3
	ORB-SLAM3 GPU	active	never	4
	ORB-SLAM3 GPU	disabled	never	3
	VINS-Fusion	—	never	—
`realistic_003`	ORB-SLAM3	active	never	3
	ORB-SLAM3	disabled	never	3
	ORB-SLAM3 GPU	active	never	3
	ORB-SLAM3 GPU	disabled	never	3
	VINS-Fusion	—	never	—

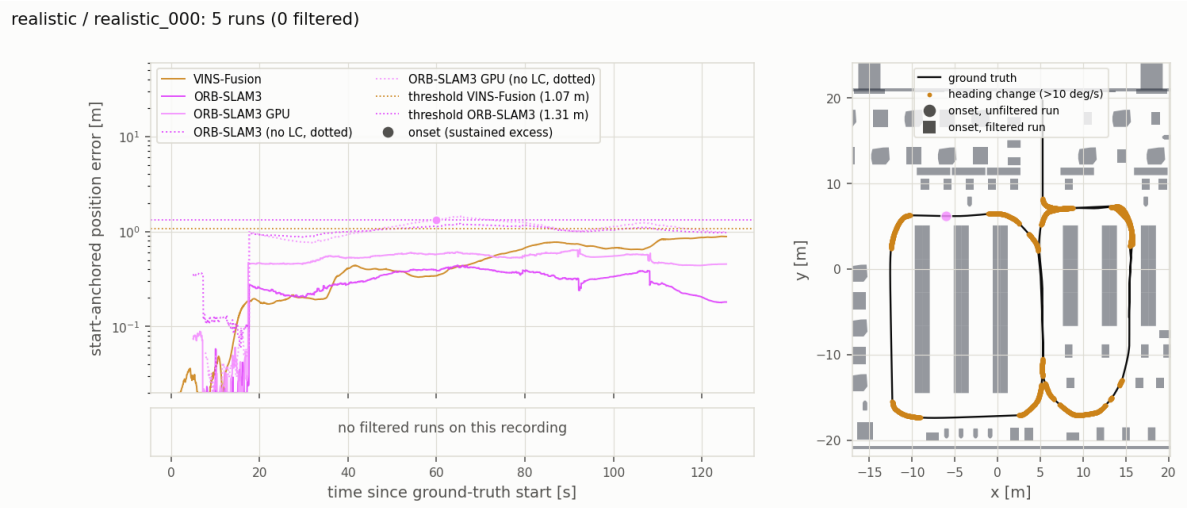


Figure 1.4: `realistic_000` (U, static), every run. Left: start-anchored planar error on a log axis, solid for loop closing active, dotted for disabled, lighter for the GPU front end, with each estimator's threshold; the dot marks the onset of sustained excess. Right: the route with heading changes in orange and every onset placed on it. The one onset is the GPU closure-free run at 60 s, a sustained 1.3 m offset on the straight, not a jump.

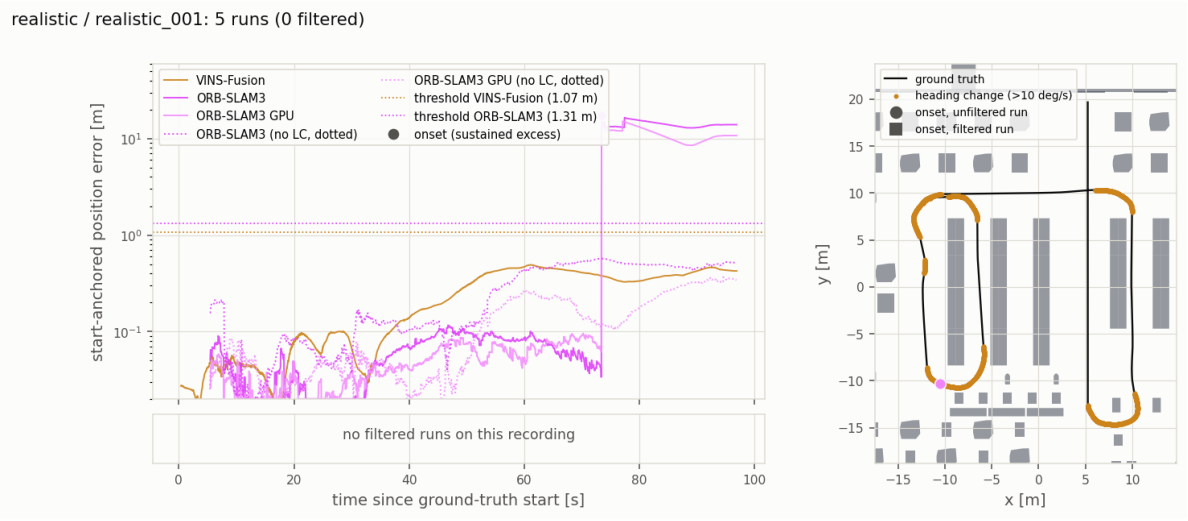


Figure 1.5: `realistic_001` (I, static), every run; panels as in figure 1.4. Both loop-closing runs step to 14 m of error at 73 s, in the heading change at 76 % of the route, at the single accepted closure; the closure-free runs and VINS-Fusion never cross.

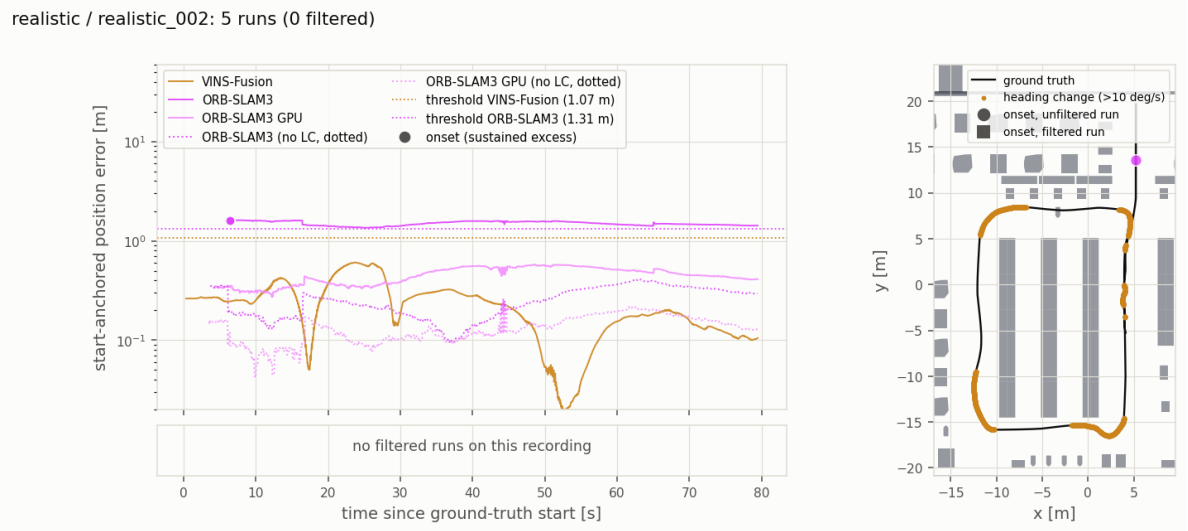


Figure 1.6: `realistic_002` (U, traffic), every run; panels as in figure 1.4. The CPU loop-closing run carries a 1.5 m offset from 6.5 s, which the GPU loop-closing run does not share; VINS-Fusion's two 0.7 m waves at 25 and 45 s lie on the aisles where the pallet jacks work.

Table 1.8: Real-time and resource measurements on the two mapping recordings, replay rate 1.0, no detector: mean ± SD over n repeats of each configuration. **Bold** marks the best mean in each column within a recording. Front-end and back-end are each run's median module latency (ORB-SLAM3: tracking and local mapping; VINS-Fusion: feature tracking and back-end optimisation); throughput is processed frames per second of sensor time; "Kept" is the processed fraction of the bag's stereo pairs; CPU is process time over wall time, in percent of one core; RSS the peak resident set. The loop-closer state is read from `loop_closing.csv`.

Recording	Estimator	LC	n	Front-end [ms]	Back-end [ms]	Throughput [Hz]	Kept [%]	CPU [%]	Peak RSS [Mi B]
`realistic_map`	ORB-SLAM3	disabled	3	42.0 ± 3.3	211.3 ± 14.8	28.3 ± 0.9	93 ± 3	379 ± 11	1732 ± 15
	ORB-SLAM3 GPU	disabled	3	**36.7 ± 1.2**	284.2 ± 14.7	**29.1 ± 0.4**	**96 ± 1**	**332 ± 7**	1794 ± 19
	VINS-Fusion	—	3	38.1 ± 3.3	**44.3 ± 5.0**	27.5 ± 2.7	91 ± 9	819 ± 10	**479 ± 15**
`realistic_01_map`	ORB-SLAM3	disabled	3	43.9 ± 1.0	220.4 ± 12.6	**27.5 ± 0.3**	**91 ± 1**	381 ± 6	1816 ± 29
	ORB-SLAM3 GPU	disabled	3	51.9 ± 3.8	405.7 ± 22.6	22.9 ± 1.4	75 ± 4	**361 ± 9**	1719 ± 21
	VINS-Fusion	—	3	**38.6 ± 0.8**	**44.5 ± 1.5**	27.4 ± 0.5	90 ± 2	814 ± 5	**497 ± 11**

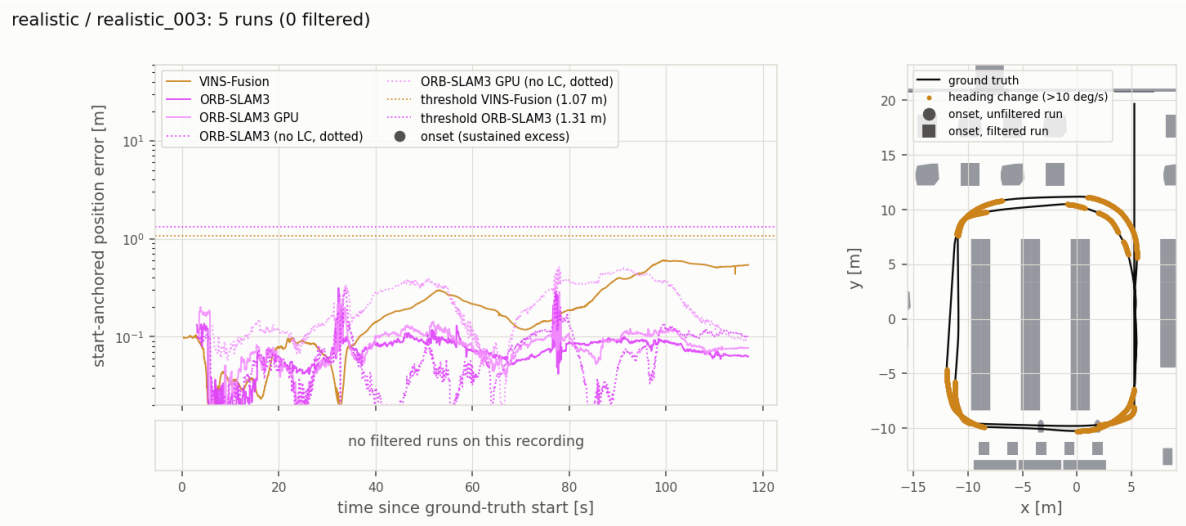


Figure 1.7: `realistic_003` (I, traffic), every run; panels as in figure 1.4. No run crosses its threshold in the plane; the 1.8 and 1.9 m ATE of the closure-free runs is vertical drift, which the single accepted closure of the loop-closing runs removes.

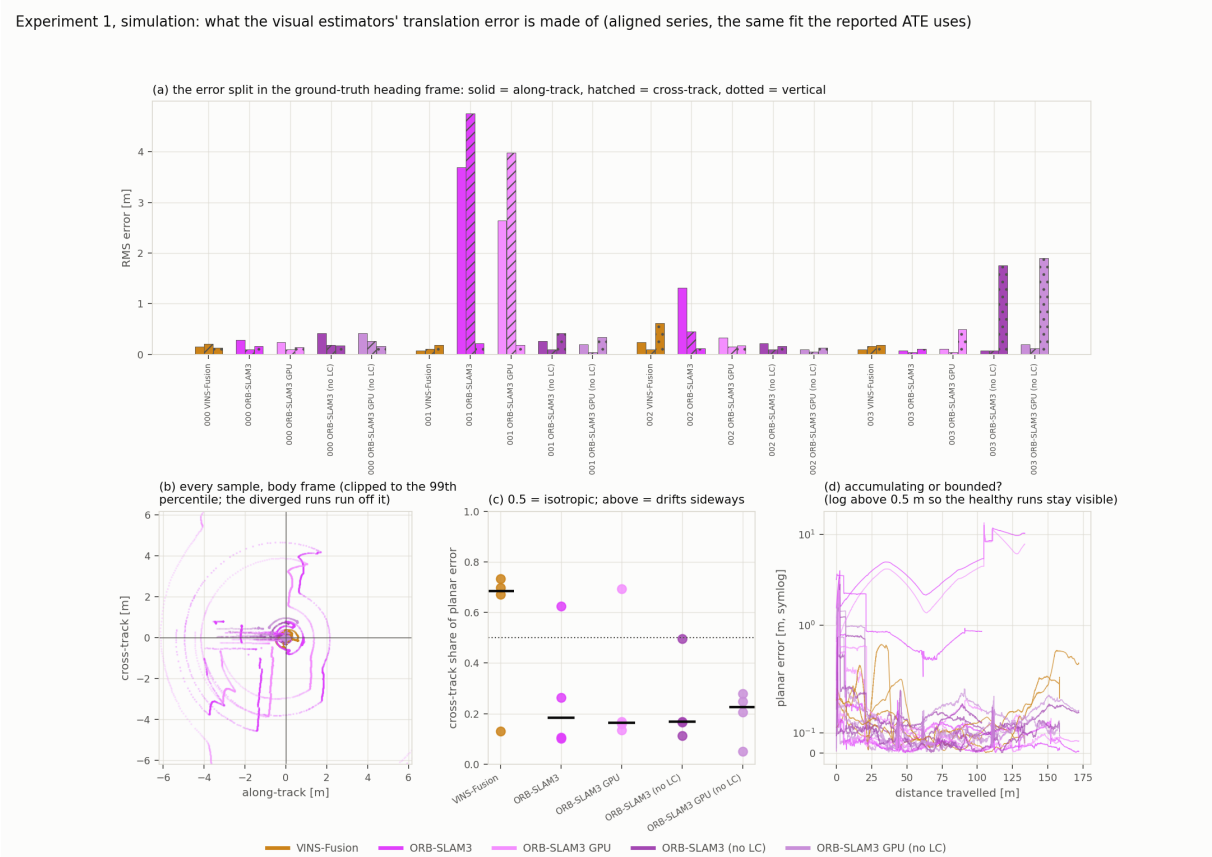


Figure 1.8: What the translation error is made of, on the aligned series the reported ATE uses. (a) the error split per run into along-track, cross-track and vertical; (b) every sample in the body frame, clipped to the 99th percentile because the five diverged runs reach metres; (c) the cross-track share of the planar error, where 0.5 is isotropic and above it the estimate drifts sideways; (d) error against distance travelled, on a symmetric-log axis above 0.5 m so that the eighteen runs that held tracking stay visible.

What each accuracy run cost. The twenty accuracy runs were measured for cost as well, on the same machine and in the same sessions, and table 1.9 reports them one row per run. They are not a substitute for table 1.8: they are single runs on four short recordings rather than three repeats on two long ones, and the divergences of table 1.4 are inside them. What they add is that accuracy and cost can be read on the same runs, which figure 1.10 draws.

The separation between the two pipelines is the one the mapping recordings show, at the same sizes. VINS-Fusion holds its state in 477–492 Mi B against ORB-SLAM3's 1029–1350 Mi B and optimises in 45–60 ms against 189–286 ms, for 707–801 % of one core against 285–448 %. Within ORB-SLAM3 the GPU front end tracks in 29.6–36.7 ms against the CPU front end's 49.2–54.9 ms and draws about a core less, at 150 Mi B more memory. These runs were not interleaved either, so they carry the same thermal caveat as table 1.8 and are quoted as a second observation of the same separation, not as an independent one.

The figure's own result is a negative one: no panel shows cost buying accuracy. Every run that diverged is ORB-SLAM3, and within each front end the diverged runs cost what the runs that held cost — with the GPU front end 31.6–36.7 ms against 29.6–36.2 ms, overlapping entirely, and with the CPU front end 52.6–54.9 ms against 49.2–51.6 ms, a gap of one to five milliseconds across three runs and two recordings, which one run per cell cannot separate from the recording. A run about to be thrown several metres by a false loop closure is not working harder beforehand, so no resource measure on this machine is a usable early warning for the failure mode this experiment actually found.

Table 1.9: Realistic world, real-time and resource measurements, one row per run. Every run is the deployment condition of section 4.1.1: replay rate 1.0, node live on the laptop, no detector running, which is why the mask column reads *off* throughout. Front-end and back-end are each run's median module latency (ORB-SLAM3: tracking and local mapping; VINS-Fusion: feature tracking and back-end optimisation); throughput is processed frames per second of sensor time; "Kept" is the processed fraction of the bag's frames, so frames lost before the node's own queue count as lost; CPU is process time over wall time, in percent of one core, so a value above 100 is more than one core; RSS the peak resident set. Accuracy for the same runs is in the accuracy table of section 4.1.1.

Recording	Estimator	Mask	LC	Atlas	Front-end [ms]	Back-end [ms]	Throughput [Hz]	Kept	CPU [%]	Peak RSS [Mi B]
`realistic_000`	ORB-SLAM3	off	active	no	51.6	272.0	21.7	72 %	445	1133
	ORB-SLAM3	off	disabled	no	50.4	258.0	22.4	74 %	438	1153
	ORB-SLAM3 GPU	off	active	no	29.6	226.8	29.9	99 %	340	1350
	ORB-SLAM3 GPU	off	disabled	no	31.4	251.7	29.9	99 %	350	1347
	VINS-Fusion	off	—	no	39.0	44.9	23.5	77 %	801	477
`realistic_001`	ORB-SLAM3	off	active	no	52.6	274.5	22.3	73 %	448	1162
	ORB-SLAM3	off	disabled	no	51.5	271.3	22.6	74 %	437	1057
	ORB-SLAM3 GPU	off	active	no	31.6	233.2	29.9	99 %	343	1349
	ORB-SLAM3 GPU	off	disabled	no	34.5	271.2	29.3	97 %	353	1264
	VINS-Fusion	off	—	no	40.8	49.9	23.8	78 %	730	487
`realistic_002`	ORB-SLAM3	off	active	no	54.9	285.8	23.4	77 %	439	1135
	ORB-SLAM3	off	disabled	no	50.8	262.4	25.4	83 %	428	1029
	ORB-SLAM3 GPU	off	active	no	30.7	189.1	30.0	98 %	285	1334
	ORB-SLAM3 GPU	off	disabled	no	34.6	262.6	29.9	99 %	327	1215
	VINS-Fusion	off	—	no	40.5	49.5	26.7	88 %	738	482
`realistic_003`	ORB-SLAM3	off	active	no	49.2	267.3	28.5	94 %	408	1141
	ORB-SLAM3	off	disabled	no	54.1	279.3	26.6	88 %	423	1157
	ORB-SLAM3 GPU	off	active	no	36.2	250.3	30.1	99 %	306	1346
	ORB-SLAM3 GPU	off	disabled	no	36.7	269.6	30.0	99 %	309	1335
	VINS-Fusion	off	—	no	46.0	60.3	26.3	87 %	707	492

How repeatable is any of this? The eighteen runs above are the only repeated runs in the simulation campaign, and scoring them for accuracy as well as for time measures something no other result in this chapter can: how much an estimate moves when nothing changes. Table 1.10 gives the spread over the three repeats of each configuration. The two estimators answer differently. VINS-Fusion repeats to 0.094 m on the U recording and 0.035 on the I, five and seven per cent of its own mean. ORB-SLAM3 does not: its three repeats span 0.089 m on the U recording and 0.277 on the I with the CPU front end, and 0.016 and 0.236 m with the GPU front end — on the I recording that is half to four fifths of the configuration's own mean, from the same binary, the same configuration file and the same bag, with the loop closer disabled in all six.

The frame counts in the same table are consistent with why. VINS-Fusion processed every frame

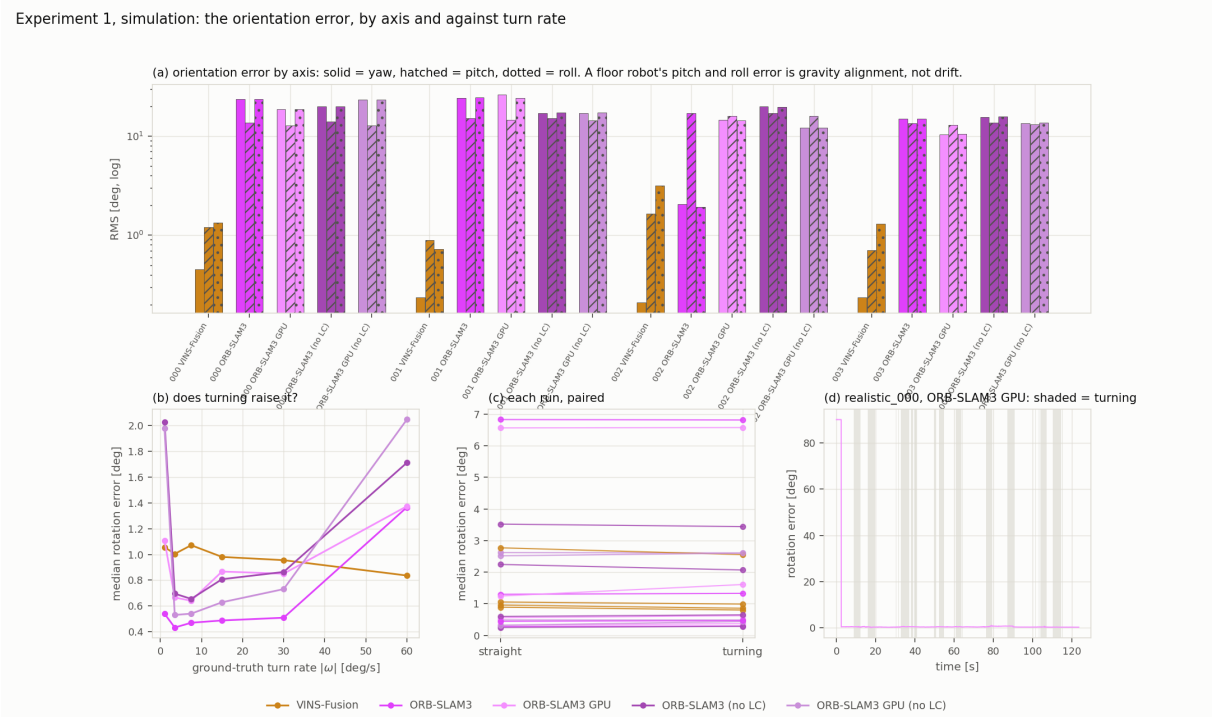


Figure 1.9: The orientation error. (a) yaw, pitch and roll RMS per run on a logarithmic axis; (b) median rotation error binned by the ground truth's own turn rate; (c) the same two medians paired within each run; (d) one run's rotation error against time, with the turning segments shaded.

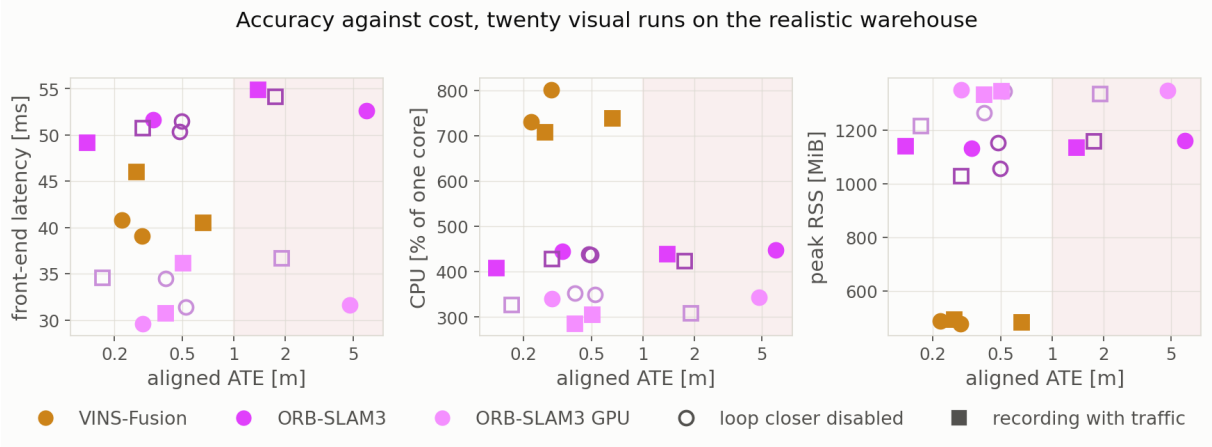


Figure 1.10: Accuracy against cost, one point per visual run on the four realistic-warehouse recordings, from tables 1.4 and 1.9. Aligned ATE is on a logarithmic axis in all three panels; the shaded band is the region above 1 m, which holds the five diverged runs. Orange is VINS-Fusion, magenta ORB-SLAM3 with the CPU front end and light magenta with the GPU front end; an open marker is a run with the loop closer disabled and a square a recording carrying traffic. Cost is the run's median front-end latency, its process time as a percentage of one core, and its peak resident set.

of both recordings in all six runs, so it saw identical input each time; ORB-SLAM3 kept 90–97 % with the CPU front end and 70–98 % with the GPU one, and which frames a run loses depends on how the laptop was scheduling at that moment. That is a consistent explanation and not a demonstrated one: a keyframe-based system with a bundle adjustment on its own thread need not be deterministic even on identical input. Separating the two would need a run at a replay rate slow enough to drop nothing, which this chapter does not have. The worst spread also belongs to the I recording, whose repeats carry the thermal confound noted above, so part of 0.277 m may be the laptop rather than the estimator.

Read against table 1.4 this sets a floor under what that table can be asked. A difference of less than about a quarter of a metre between two ORB-SLAM3 arms is inside the spread measured here and should not be read as an effect; this is the quantity the bold marking of that table was hedged against, now measured. Two of the chapter's findings survive it and one is strengthened. The five divergences are 1.39–6.02 m, five to twenty times the floor, and the false closure that causes the largest of them is reproduced by both front ends on the same frame. The claim that the front end changes throughput and not accuracy is supported rather than weakened: the CPU and GPU arms differ by less than this floor on every recording, which is what having no effect looks like. What should not be carried further is any ordering of two ORB-SLAM3 arms that rests on a gap smaller than it.

Table 1.10: Run-to-run spread over the three repeats of each configuration on the two mapping recordings, the only repeated runs in the simulation campaign; the same eighteen runs as table 1.8, scored for accuracy. Every run has its loop closer disabled, read from `loop_closing.csv`. "ATE" is mean ± SD over the three repeats and "spread" the largest difference between two of them; "Max" is the largest error after the initialisation transient, on the convention of the accuracy table of section 4.1.1. The three repeats of a configuration were run back to back rather than interleaved, so a drift across them is confounded with the laptop's thermal state (section 4.1.1); the Kept column is where that would show.

Recording	Estimator	n	ATE [m]	spread [m]	Max [m]	ATE/path [%]	Kept
`realistic_map`	ORB-SLAM3	3	0.189 ± 0.045	0.089	0.748 ± 0.267	0.05 ± 0.01	0.94 ± 0.03
	ORB-SLAM3 GPU	3	0.144 ± 0.008	0.016	0.345 ± 0.033	0.04 ± 0.00	0.96 ± 0.01
	VINS-Fusion	3	1.915 ± 0.049	0.094	3.617 ± 0.239	0.51 ± 0.01	1.00 ± 0.00
`realistic_01_map`	ORB-SLAM3	3	0.348 ± 0.149	0.277	1.517 ± 0.749	0.09 ± 0.04	0.91 ± 0.01
	ORB-SLAM3 GPU	3	0.433 ± 0.118	0.236	2.691 ± 0.338	0.11 ± 0.03	0.76 ± 0.04
	VINS-Fusion	3	0.491 ± 0.017	0.035	5.088 ± 0.167	0.13 ± 0.00	1.00 ± 0.00

Limitations, each stated once. *n:* one run per cell on the four accuracy recordings, in both families; the three repeats of each mapping-recording configuration (table 1.10) are the only within-configuration spread the campaign measures, and they are on different routes. *Equal handicap, not equal effort:* the shared inertial model was fitted to neither family. *Idealised wheels:* simulated encoders with slip under 1 %, the proprioceptive estimators' best case. *Route and traffic confounded:* no two recordings share a route, so traffic is a condition of a recording and not a tested factor. *Three-dimensional ATE:* the metric charges vertical drift to a floor robot; the planar onset table is the complement, not a replacement. *Deployment rate on one laptop:* frame retention and every latency figure are properties of this machine at rate 1.0, and the mapping-bag repeats were not interleaved between front ends. *Run bookkeeping:* one directory held a run on another recording for a day; the manifest caught it, and the launch should refuse an output directory whose name disagrees with its dataset.

[Resolved: `realistic_002/loopclosing_gpu`] Earlier on 6 November this run was reported as closure-free from its `loop_closing.csv` and queried. Its manifest shows the real cause: it was launched on the `realistic_003` bag with `realistic_002`'s output path, so it was scored against the wrong truth (4.94 m, 48.9°, 139 % of the recording's frames processed). It is set aside, not deleted; the configuration was re-run on the right bag the same evening and every table here carries that run.

[Open: the false closure is diagnosed from its correction, not from the images] The 14 m correction and the one-aisle keyframe separation identify the `realistic_001` closure as false, and both front

ends agree; the matched image pair has not been inspected, and the two April-style guards that would catch it (a covisibility-consistency window, or a maximum correction against the inertial drift bound) are untested. Until one is, the finding is a diagnosis, not a fix.

Summary. Two families were scored on the same four recordings. Among the non-visual estimators IMU-only dead reckoning drifts by 12–23 m and wheel odometry by 1.0–4.8 m of aligned ATE; the EKF, run on the visual systems' own inertial noise as the design requires, holds 0.68–2.80 m, while the same filter on a measured noise model would hold 0.21–0.58 m. Among the visual ones VINS-Fusion holds 0.22–0.66 m with no jump and is worst on the U layout with traffic; ORB-SLAM3 holds 0.14–0.52 m in eleven runs of sixteen, and its five divergences are the loop closer's: a false closure on the I layout's repeated aisles that puts both front ends 4.8–6.0 m off, a vertical drift on the same layout with traffic that only the closer corrects, and one initialisation offset. Neither pipeline loses tracking on any recording, with or without traffic, and the front end changes throughput (97–99 % of frames kept against 72–94 %) and not accuracy. Across the families the EKF at the shared handicap loses to VINS-Fusion on every recording and the measured-noise diagnostic beats it on three of four, so on these routes the noise model is worth more than the choice of estimator. Against the project's target of 0.05 m ATE no estimator qualifies; against the 1.5 m RPE bound at 1 s, all but the IMU do. On cost, VINS-Fusion optimises five to nine times faster than ORB-SLAM3's local mapping and in a quarter of the memory, for more than twice the processor time.

Future plan. Fit the visual systems' inertial noise to this platform rather than copying a file, since the slot shows the noise model is worth more than the estimator; feed the VIO pose into the EKF as a third measurement; add a correction bound to the loop closer and test it on `realistic_001`; repeat the mapping-recording runs interleaved between all three configurations from a cool start, so that the thermal state is shared rather than confounded with the configuration; and record a second route in each layout so that traffic and route are separable.

4.1.2 Real robot

Status: partial. Four real recordings with an independent planar reference (`dataset_realv6_000`, `dataset_realv7_000` to `_002`). A non-visual baseline exists: wheel+gyro dead reckoning with the feedback speed scaled by a factor fitted once against the tracker on `realv7_000` and held on the other three. The visual arm is ORB-SLAM3 in four configurations on each recording, sixteen runs, all made with the fork's patch-15 initialisation switches and scored with no repair applied; they give 0.404–0.480 m on `realv6_000`, 0.251–0.268 on `realv7_000`, 0.442–1.278 on `_001` and 0.172–0.198 on `_002`. Across the held recordings the two families are in the same accuracy class, the baseline leading on one, tying on one and losing the third. The EKF of the simulation slot cannot be formed, because the rig publishes no steering measurement; VINS-Fusion has no valid run on any recording; and every value is provisional until the marker-to-IMU lever arm is measured.

Objective. The real-robot half of the simulation question: do the visual pipelines track the robot against an independent reference, and does a non-visual estimator match them on the same recordings.

Process. Two groups of runs share these recordings.

Non-visual. The bags carry `/feedback/velocity` and `/feedback/steering` (50 Hz, `std_msgs/Floa` and the WitMotion yaw sensor on `/hwt101ct_yaw_publisher` (200 Hz). The velocity is in a firmware unit, not metres per second; the steering is a command echo that takes two values (0 and −30) on every recording and so carries no yaw rate; the yaw sensor's orientation field is identity and its rate

on the y axis is the live gyro (the sensor is mounted with y vertical: that axis integrates to 6.29 rad over `realv7_000`, whose tracker heading sweeps 6.3 rad; x and z integrate to under 0.05 rad).

`script/real_wheel_baseline.py` integrates heading from that gyro with its rest bias removed (3–7 s at rest at the head of each bag) and position from the scaled speed. The scale k, the gyro axis and its sign are fitted on `realv7_000` alone by minimising the planar ATE after rigid $SE(2)$ alignment with scale fixed (a wrong sign mirrors the path, which a proper rotation cannot align), then held: the fit recording's own score is a calibration, the three held scores are the result. The fitted $k = 0.325$ agrees with the plain path-length ratio of each `realv7` bag (0.323 to 0.329). The time offset between tracker and bag is the wheel-speed cross-correlation of `script/eval_real_gt.py` ($r = 0.93$ to 0.99 on `realv7`, 0.61 on the sparse `realv6` export), scoring is that script's planar procedure at the tracker's own samples, and the one off-path `realv6` sample is removed by the rule the realv6 analysis fixed.

The rig's wheelbase, 0.60 m, is recorded and unused: with no steering measurement there is nothing to put through a bicycle model, so the yaw-rate observation that gives the simulation EKF its bias state does not exist, and that filter would reduce to this integration.

Visual. The bag was replayed into each node at real time on the afternoon of 3 November 2026; no detector ran. ORB-SLAM3 used `config/wil/stereo_imu.yaml` with the repaired library (fork patches 11–14b, node and library built alike), the 3 November stereo extrinsic and `IMU.NoiseGyro` 0.1 with `IMU.NoiseAcc` 1.0, ten times the values that had been copied from the VINS file, in four configurations: GPU or CPU feature extractor, loop closing on or off. VINS-Fusion used the same file with the gravity norm of an August recording. Earlier runs on this bag were overwritten by these and are not reported. The optical tracker export was scored with the real-robot procedure of Experiment 1.

Dataset. 220 s, 6 601 stereo pairs at 30 Hz, RealSense IMU at 200 Hz on the inherited topic name, wheel-speed feedback, a figure-of-eight of about 27 m at a median 0.15 m s^{-1} on the Gensurv floor, no scripted moving objects. Tracker export: 277 samples, median spacing 0.6 s, 66 gaps above 1 s, one sample 5.2 m off the path at 184.9 s.

Datasets. The four real recordings that carry an independent planar reference: routes of 22–37 m over 97–219 s. Sources: `output/compare/real/exp1-wheel/` (`wheel_metrics.csv`, `manifest.json`), estimates under `output/output_wheel/real/`.

Reporting rules, visual arms. `script/eval_real_gt.py`: rigid SE(2) alignment with scale fixed, lever arm set to zero and unmeasured, every value provisional. The export fails the tool's gap and step rules, so the tool labels each run *failed (ground truth invalid)*; the values below are diagnostics. All are scored with the one off-path sample removed, an edit recorded here and applied nowhere else; table 1.13 shows what that sample does to one run. The bag was not on the machine, so the time offset is pinned to the value the wheel-speed cross-correlation found on 2 November and the manifests record the substitution. Sources: `output/compare/real/realv6_1103pm_*/`, `output/compare/real/realv6-insights/`, run directories under `output/output_orb/real/dataset_realv6_000/` and `output/output_vins/real/dataset_realv6_000/`.

Non-visual baseline

Results. Table 1.11 and figure 1.11. On the fit recording the baseline reads 0.06 m ATE with a path ratio of 0.99, which is the calibration. Held, it reads 0.26 and 0.22 m on `realv7_001` and `_002` with path ratios of 0.99 and 1.01, and 0.46 m on `realv6_000` against its sparse export, with no jump and a relative error at 1 s of 0.01–0.02 m.

Table 1.11: Wheel+gyro dead reckoning on the real recordings against the optical tracker: the feedback speed scaled by k (fitted on `realv7_000` by minimising planar ATE, held on the others), heading from the integrated gyro with its rest bias removed, rigid $SE(2)$ alignment with scale fixed, lever arm zero and unmeasured. ATE at the tracker's samples; path ratio is estimated over true path length.

Recording	Role	k	Offset [s] (r)	Samples	ATE RMSE [m]	ATE max [m]	Path ratio	$\mathrm{RPE}_{1\,\mathrm{s}}$ [m]
`realv7_000`	fit	0.325	-2.57 (0.99)	491	0.057	0.11	0.990	0.008
`realv7_001`	held	0.325	-3.77 (0.98)	632	0.259	0.47	0.988	0.021
`realv7_002`	held	0.325	-2.60 (0.93)	499	0.219	0.27	1.006	0.008
`realv6_000`	held	0.325	-4.14 (0.61)	233	0.463	0.70	1.041	0.015

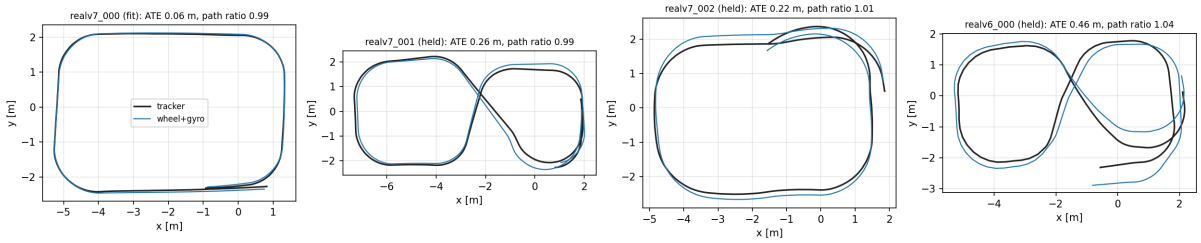


Figure 1.11: Wheel+gyro dead reckoning (blue) against the tracker (black) after rigid planar alignment, one panel per recording; the first is the recording the speed scale was fitted on.

Analysis. The comparison this slot exists for now has a number on both sides, and it does not favour vision. On the same recordings the best ORB-SLAM3 configuration reads 0.44 m on `realv7_001`, 0.47 m on `realv7_002` and 0.46 m on `realv6_000` (the real-robot section below, after the scorer's frame repair); the scaled dead reckoning reads 0.26, 0.22 and 0.46 m, with path ratios within 4 % of one where the visual runs read 5–15 % long.

Three things bound that reading. The scale is a fitted constant, not a measured one, so the baseline has had one parameter tuned against the truth it is scored on, on a different recording; a second calibration route, the wheel circumference and the gear ratio, would remove that. The routes are 22–37 m with the robot at 0.2 m s^{-1}: a gyro drifting at the measured rest bias of 0.003–0.006 rad s^{-1} accumulates 0.3–1.3 rad over a run if the bias is not removed and under 0.1 rad after the rest-window removal, and on a longer route that term grows while the visual error need not. And the lever arm is zero for both sides. Against the project's targets no run approaches 0.05 m except the calibration itself; every run is inside the 1.5 m RPE bound.

[Open: the realv6 scale] On `realv6_000` the path-length ratio gives $k = 0.43$ where the three `realv7` bags give 0.32 to 0.33, yet the held $k = 0.325$ scores a path ratio of 1.04 on the matched samples. The export is at 1.7 Hz and fails the validity rules, and the recording predates `realv7` by three days; whether the firmware unit changed or the sparse export misstates the path length is not decided, and the `realv6` row is a diagnostic.

Visual estimators

Results. ORB-SLAM3 follows the whole 218 s route in every configuration: 0.52 and 0.46 m with the GPU extractor (loop closing on and off), 0.62 and 1.03 m with the CPU extractor, which drops two thirds of the frames at real time (table 1.12, figure 1.12). VINS-Fusion tracked 18 s before 217 IMU-gap resets took over (table 1.15) and has no entry in the accuracy table.

Table 1.12: ORB-SLAM3 on `dataset_realv6_000` against the cleaned tracker truth, 3 November 2026 afternoon series: repaired library, replay rate 1.0, 3 November stereo extrinsic, `IMU.NoiseGyro` 0.1 and `IMU.NoiseAcc` 1.0, GPU or CPU feature extractor, loop closing (LC) on or off. Aligned planar ATE after a rigid SE(2) fit with scale fixed; RPE at 1 s translational; path ratio = estimate / truth; resets = `local_mapping_bad_imu` map resets; dropped = camera frames the node's queue discarded of about 6 480 received. Lever arm tracker-marker to IMU unmeasured (zero applied); time offset pinned to −4.14 s; all values provisional under the scorer's validity rules.
textbfBold marks the lowest ATE of the five runs; with one run per configuration that records which run came lowest, not that the configuration is better.

Run	ATE [m]	median [m]	max [m]	$RPE_{1\,s}$ [m]	path ratio	resets	dropped
ORB-SLAM3, GPU, LC on	0.52	0.44	1.11	0.057	0.85	2	2420
ORB-SLAM3, CPU, LC on	0.62	0.48	1.38	0.110	0.88	2	4363
ORB-SLAM3, GPU, LC off	**0.46**	0.36	0.99	0.056	0.87	0	3201
ORB-SLAM3, CPU, LC off	1.03	0.89	1.83	0.223	0.96	3	4204

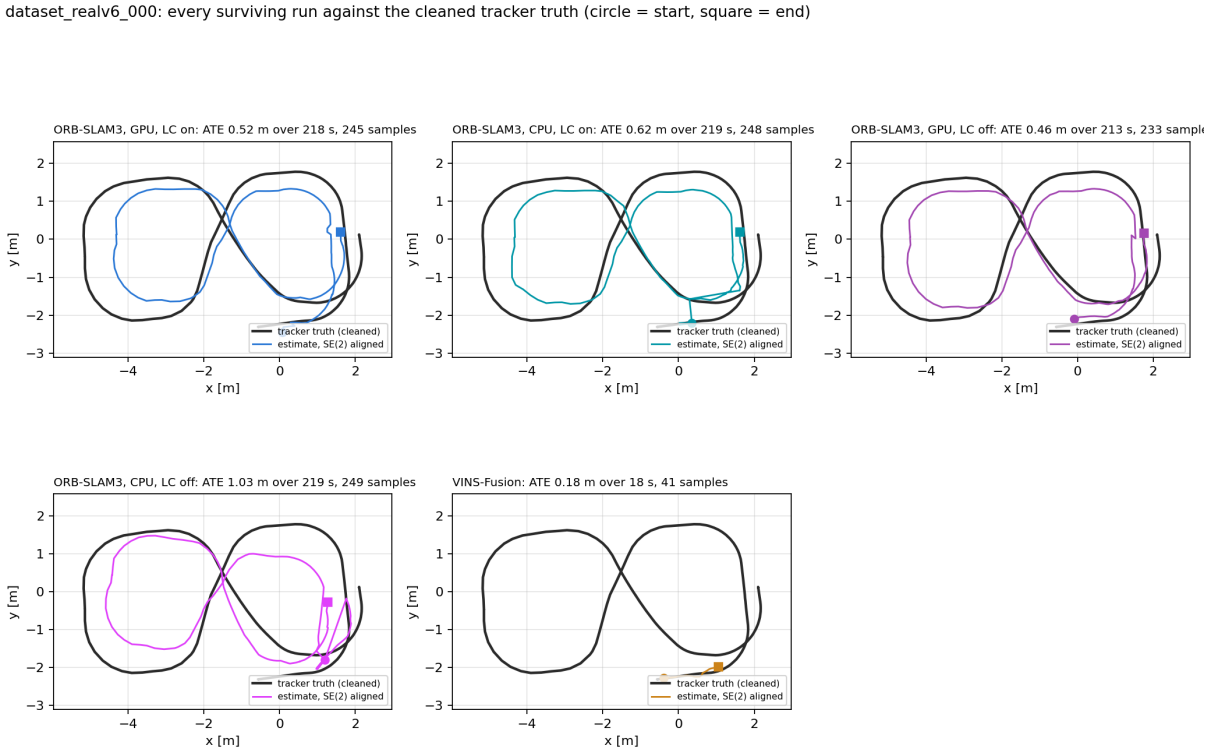


Figure 1.12: Every run against the cleaned tracker truth after rigid SE(2) alignment, one panel per run: ORB-SLAM3 with GPU or CPU extractor and loop closing on or off, and VINS-Fusion's 18 s.

Table 1.13: The same ORB-SLAM3 run (GPU, LC on) scored against the tracker export as delivered and with one sample removed (the 5.2 m excursion at 184.9 s). The time offsets are the values the wheel-speed cross-correlation found on each export on 2 November (peak correlation 0.10 raw, 0.61 cleaned), applied here as pinned values. The cleaned truth still fails the gap rule (66 gaps above 1 s, tracker at 1.7 Hz).

Truth	GT path [m]	offset [s]	ATE [m]	median [m]	path ratio
as delivered	36.8	4.38	1.174	0.851	0.65
one sample removed	27.1	-4.14	0.519	0.437	0.85

Analysis. The front end is what separates the runs. The GPU extractor drops a third to a half of the frames and holds 0.5 m; the CPU extractor drops two thirds and loses a factor of two, one of its runs with a 2 m vertical excursion the tool flags as implausible and the other with three resets and a half-second of lost tracking. Loop closing changes nothing that the front end does not change more. The remaining resets are an initialisation effect, two or three `local_mapping_bad_imu` map resets inside the first 13 s and none after (zero in the GPU run without loop closing); after them the error is a slow wave between 0.05 and 1 m, not an accumulating drift (figure 1.13), and it does not rise while turning in any run (table 1.14). The estimates are a uniform 12–15 % smaller than the tracker's path, which the unmeasured lever arm, a stereo-baseline error or a tracker scale error could each explain. One tracker sample decides the headline: scored against the export as delivered, the GPU run reads 1.17 m with a path ratio of 0.65, because that sample adds 10 m of false path and dominates the time alignment (table 1.13).

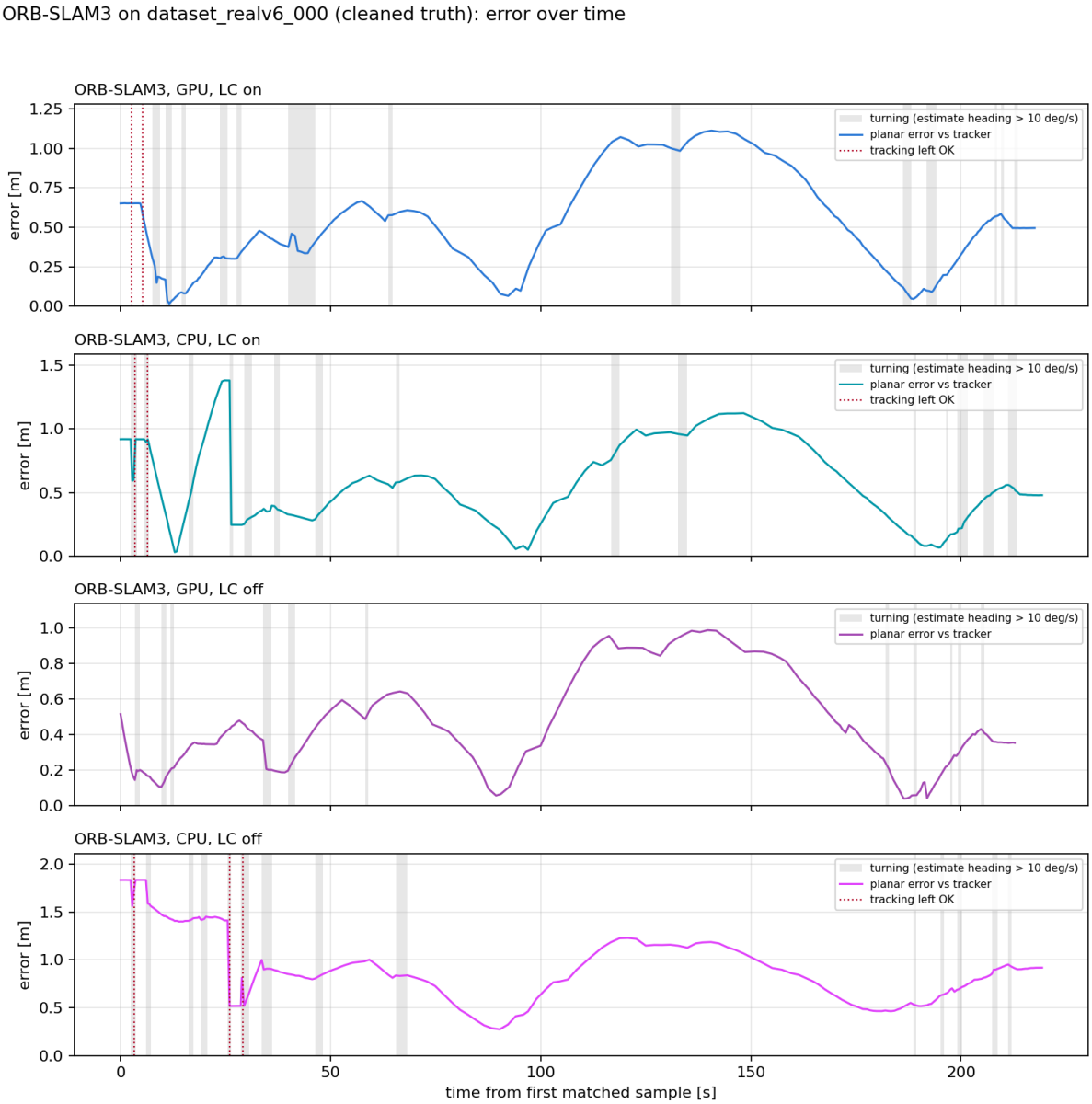


Figure 1.13: ORB-SLAM3 planar error over the run, one panel per configuration; grey bands are turns, dotted red the initialisation resets.

The setting was reached by elimination. With the IMU noise densities copied from the VINS file the estimator was unstable on this recording, running away in its velocity state at a different place in each run;

Table 1.14: ORB-SLAM3 on `dataset_realv6_000`, cleaned truth: growth rate of the planar error while the aligned estimate's heading changes faster than $10\,^\circ\,s^{-1}$ and on straights, with the time in each state. Positive = error growing. One run per configuration, so no test is possible.

Run	turning [mm/s]	straight [mm/s]	time turning [s]	time straight [s]
ORB-SLAM3, GPU, LC on	-11.2	-0.3	23	171
ORB-SLAM3, CPU, LC on	-29.8	-1.4	23	162
ORB-SLAM3, GPU, LC off	4.9	-2.1	11	185
ORB-SLAM3, CPU, LC off	-49.9	-0.9	20	182

six out-of-bounds reads in the library were fixed without removing it, every other candidate was varied without removing it, and raising the IMU noise densities tenfold removed it in every run since. Those earlier runs were overwritten and carry no number here; the search is in the ORB-SLAM3 engineering record.

VINS-Fusion's failure is a delivery defect. The node received 3 187 of 6 601 frames but used 6 498 of the 43 494 IMU messages it received, reset 217 times on `imu_gap`, and stopped after 19 s; behind that defect the configuration still fixes a gravity norm of 9.2642 against the $9.878\,m\,s^{-2}$ this IMU reads at rest.

Table 1.15: VINS-Fusion on `dataset_realv6_000`, 3 November 2026 (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm still 9.2642 against the $9.878\,m\,s^{-2}$ this IMU reads at rest). The scored span is the 18 s it tracked before its last reset and is not a result.

Quantity	Value
camera frames received (bag: 6 601)	3187
IMU messages received / samples used	43494 / 6498
`imu_gap` resets	217
poses written / span of poses	286 / 19 s
tracker samples matched	41 of 43 in span
ATE over the tracked span	0.18 m (not a result)

Second recording set, realv7. Three further recordings of 3 November 2026 (`dataset_realv7_000`, `_001`, `_002`: 21.6, 31.7 and 23.4 m over 97, 134 and 104 s) have tracker exports that pass every validity rule (5–6 Hz, no gap, no step). Each carries the same four ORB-SLAM3 configurations and one VINS-Fusion run. As written in the live log (table 1.16, figure 1.15), a plane fitted to the ORB-SLAM3 estimate lies within 10° of horizontal in three runs of twelve and at 88–90° in five, every `000` run among them, with the plane normal exactly a body axis in every run; scored as written, those nine runs are not scorable.

The per-frame diagnostics give the mechanism. The IMU initialised 3–4 s after the first pose. The upstream rule that resets a map with less than 2 cm between consecutive keyframes in the first ten seconds after the initialisation then discarded it, the robot being still at rest. The new map starts in the camera frame of the reset instant, and in it the re-initialisation was starved: upstream attempts it only with an empty keyframe queue, which a keyframe every 0.25 s and 250 ms of map-point creation per keyframe never allowed. The IMU re-initialised after 37–92 s in seven runs, never in two, and early (6.7 s) in the one run that was level. Until then the estimate is stereo-only in a map whose vertical is the camera's y axis, and the late re-initialisation rotates the map under the already-written poses (the 1.9–6.8 m steps).

A scorer-side repair was written while the cause was being found: it rotates the pre-initialisation poses by the rotation measured at the initialisation frame and levels each map's fitted plane, both without using the truth. It is kept for the record and is not used for any result below, because the defect it worked around is now fixed in the estimator; the one figure it produced that is still worth quoting is that on `000` it recovered 0.252–0.275 m from runs whose estimates were lying on their side.

The fork's patch 15 adds two switches that attempt the initialisation with a queued backlog and skip

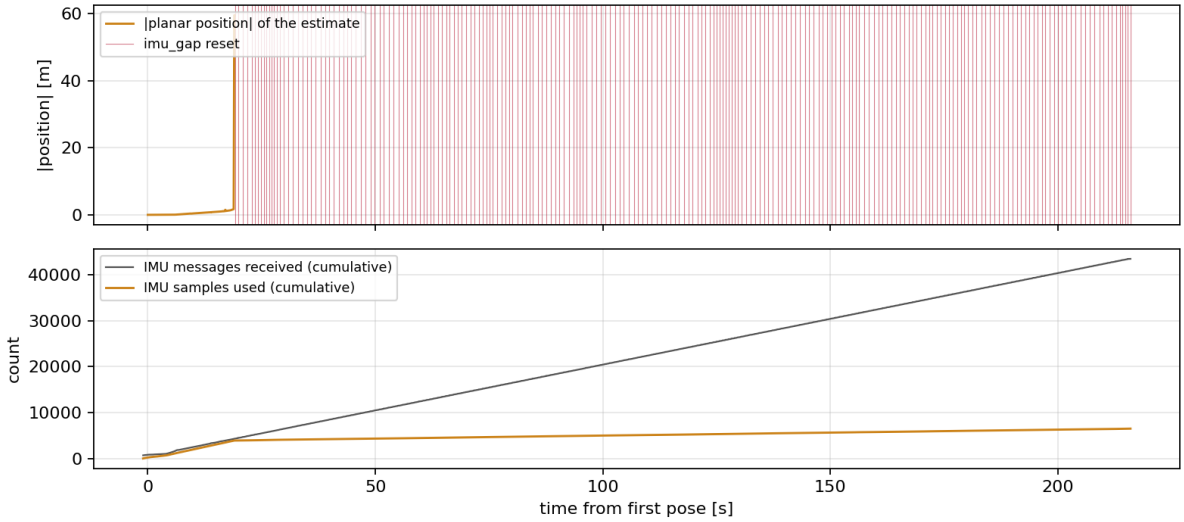


Figure 1.14: VINS-Fusion: distance from origin with every `imu_gap` reset marked, and IMU messages received against samples used.

the low-motion reset for stereo. On 11 November 2026 all twelve runs were re-made with both active, and table 1.17 reports them scored *raw*: neither of the two repairs above was applied, which is the point of the table.

The upstream behaviour is gone from every run. The IMU initialised at 2.8–4.7 s and was retained; thirteen of the sixteen request no map reset at all and the other three request one for a timestamp jump, not for insufficient motion. The plane of the estimate now lies 0.2–2.8° from horizontal in every run against the 5–90° of table 1.16, and the vertical range falls from as much as 9.6 to at most 1.00 m. Under the scorer's own rules seven `realv7` runs are now **completed** where none was before and the other five are **partial**; the four `realv6` runs remain rejected, not for anything the estimator did but because that recording's tracker export fails the validity rules.

The accuracy claim is narrower than the defect claim, and the three recordings disagree. Set against the same configuration under upstream initialisation *after both repairs* — each estimator at the best obtainable from it — patch 15 matches it on `realv7_000` (0.251–0.268 against 0.252–0.275 m), is mixed on `_001`, where two runs improve and two worsen and one of those reaches 1.278 against 0.817 m, and beats it on `_002` (0.172–0.198 against 0.472–2.115 m, every one of the four better than anything the repair obtained there) and on `realv6_000`, where all four improve (0.404–0.480 against 0.457–1.028 m) and the two CPU runs improve most. The defensible statement is therefore not that patch 15 is more accurate. It is that patch 15 removes the need for the repair: it reaches on the raw estimate what the stock estimator reaches only after its frame bookkeeping has been corrected offline, and on the one recording the repair handled worst it reaches considerably more. The 0.172–0.198 m runs of `_002` are the most accurate real-robot ORB-SLAM3 figures in this report.

[Open: what the improvement is owed to] Two things differ between the two sets of runs, not one. Patch 15 is the first; the second is that `IMU.NoiseGyro` and `IMU.NoiseAcc` were inflated tenfold on 3 November, and no artefact records which values that afternoon's runs used, because `run_metadata.csv` stores the configuration's path and not its contents. The loss of the tilt is patch 15's: the upstream rule tests keyframe spacing below 2 cm within 10 s and cannot be influenced by an inertial noise density. Any part of the accuracy difference is not separable on the present evidence. One control run, the same bag at today's noise with both switches returned to upstream, would separate them and has not been made.

[Provenance: the 3 November estimates are gone] The re-runs overwrote `output/output_orb/real/dataset_realv7_00*/` in place, so the pre-patch-15 estimates of all three recordings no longer exist and neither table 1.16 nor figure 1.15 can be regenerated; the frozen copies beside the generator are the only record of the behaviour this section reports as corrected. A run directory should be written under a name that carries its configuration, so that a re-run cannot overwrite the evidence for the behaviour it replaces.

VINS-Fusion diverges on all three with the IMU stream intact (zero resets) and 40–45 % of the frames received (table 1.18); its gravity norm has since been set to the measured 9.886 m s^{-2}.

Table 1.16: ORB-SLAM3 on the three `dataset_realv7` recordings, 3 November 2026 afternoon, same setting as the realv6 runs: the plane of each estimate as written in the live log, before any repair. "Tilt" is the angle between the plane fitted to the estimate's 3-D positions and the horizontal; "off-plane" the rms distance of the positions from that plane. A floor robot's estimate should be horizontal: a tilt near 90° means the poses were written while the map was not gravity-aligned (IMU not initialised; the vertical is then the camera's *y* axis). The scores are in the patched runs of the real-robot results table.

Recording	Run	tilt [°]	off-plane [m]	z range [m]	resets
`realv7_000`	GPU, LC on	89.1	0.35	6.87	3 s
`realv7_000`	GPU, LC off	89.5	0.02	6.98	3 s
`realv7_000`	CPU, LC on	88.8	0.41	6.88	5 s
`realv7_000`	CPU, LC off	87.9	0.58	6.79	4 s
`realv7_001`	GPU, LC on	88.8	0.05	9.64	4 s
`realv7_001`	GPU, LC off	6.1	1.12	5.77	3 s, 5 s
`realv7_001`	CPU, LC on	1.5	0.07	0.38	4 s
`realv7_001`	CPU, LC off	13.0	1.05	5.23	5 s
`realv7_002`	GPU, LC on	37.7	0.87	4.08	3 s, 49 s, 59 s
`realv7_002`	GPU, LC off	59.7	0.83	4.80	49 s
`realv7_002`	CPU, LC on	5.2	0.73	2.94	none
`realv7_002`	CPU, LC off	21.7	0.84	7.03	none

[Sparse reference] The tracker ran at 1.7 Hz rather than its nominal 7 Hz on `realv6`, and the validity rules were written for the faster export. Either the export has to be re-captured at full rate or a rule for sparse references has to be written before any value from this recording is quoted outside a diagnostic.

Summary. On the four real recordings the two families are in the same accuracy class. Wheel+gyro dead reckoning, with one constant fitted on `realv7_000` and held, localises the robot to 0.26 and 0.22 m on the other two `realv7` recordings and 0.46 m on `realv6`; ORB-SLAM3, with the upstream initialisation defect patched out and scored raw, reads 0.44, 0.17 and 0.40 m on the same three. The baseline leads on one, ties on one and is beaten on the third. The cause of the IMU staying uninitialised for most of nine of the original twelve `realv7` runs — an upstream map reset at the initialisation and a starved re-initialisation — is diagnosed, patched in the fork and now exercised on every run; what the patch is shown to do is remove the need for the scorer's frame repair, not to improve accuracy, which it matches on one recording, splits on a second and improves on the other two. VINS-Fusion has no valid run on any recording, through two delivery and configuration defects that are diagnosed and in part corrected. The marker-to-IMU lever arm is unmeasured, so every figure here is provisional, and `realv6`'s tracker export fails the validity rules written for the faster one, so its rows are diagnostics.

Next action. Run one `realv7` recording with both initialisation switches returned to upstream at the current inertial noise, which is the control that separates patch 15 from the tenfold noise change made the same week; re-run VINS-Fusion with the corrected gravity norm; measure the marker-to-IMU lever arm, which every real figure in this chapter is provisional on; and replace the fitted wheel-speed constant with a measured one. Write each run under a directory name that carries its configuration, so that no re-run overwrites the evidence it replaces.

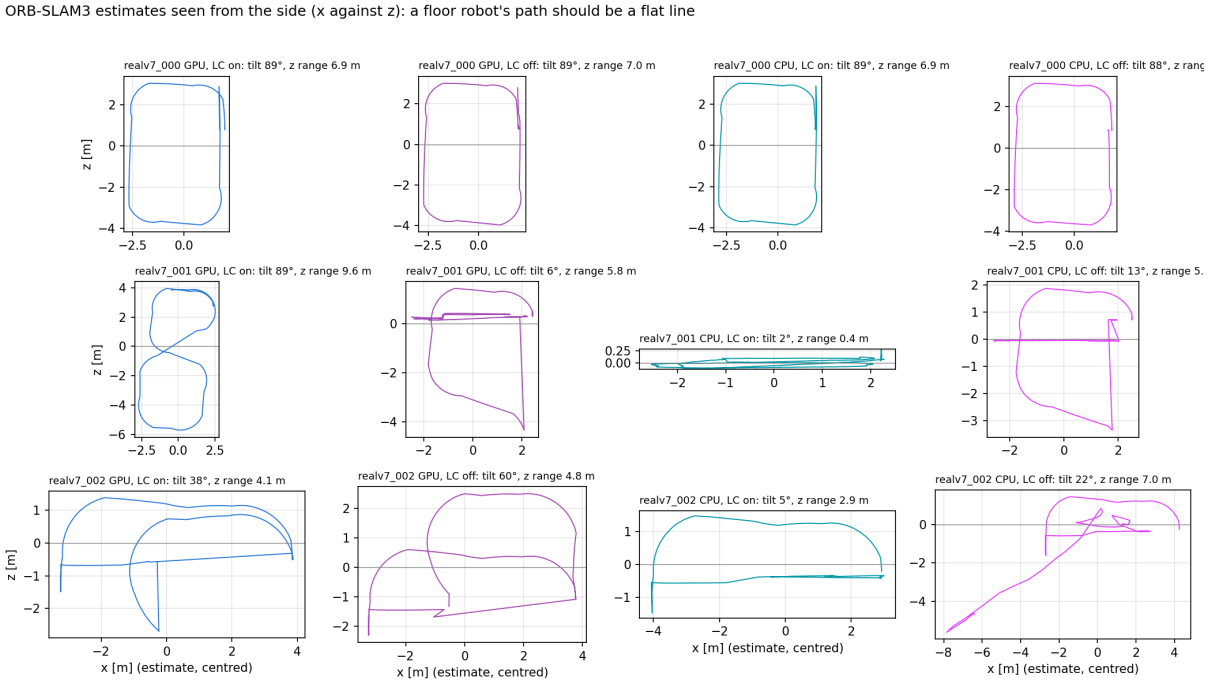


Figure 1.15: The twelve `realv7` ORB-SLAM3 estimates as written in the live log, seen from the side (*x* against *z*, centred); a floor robot's path should be a flat line. This is the defect that fork patch 15 removes; the estimates it produces instead are figure 1.16.

Table 1.17: Every real-robot ORB-SLAM3 run on the four recordings that carry an independent planar reference, with fork patch 15 active (`LocalMapping.resetOnLowMotion` 0, `LocalMapping.initWithBacklog` 1) and scored *raw*: neither `-join-imu-init` nor `-level-plane`, the repairs every other `realv7` figure in this chapter depends on. "Tilt" and "z range" are measured the same way as in table 1.16, so the two are comparable. "ATE before" is the same configuration on the same recording on 3 November, scored raw. Front-end identity was confirmed from the median tracking latency (GPU 41–50 ms, CPU 104–113 ms), not from the run name. "Stock + repair" is the same configuration on the same recording on 3 November, under upstream initialisation and, for `realv7`, only after the scorer-side frame repair that has since been retired: it is the fair comparison, each estimator at the best obtainable from it. For `realv6_000`, which never showed the tilt and needed no repair, "before" is the same run scored against the same cleaned tracker export without the switches. Status is the scorer's: it rejects a run whose estimate spans more than 1 m vertically, and it rejects every `realv6` run because that recording's tracker export is 1.7 Hz and fails the validity rules written for the 7 Hz one, so its rows are diagnostics. Lever arm still unmeasured, so every value is provisional.

Recording	Run	IMU init [s]	resets	tilt [°]	z range [m]	Kept	ATE [m]	Before [m]	status
`realv6_000`	GPU, LC on	3.1	0	1.1	0.27	0.58	0.478	0.519	failed
	GPU, LC off	4.5	1	1.2	0.33	0.51	0.404	0.457	failed
	CPU, LC on	3.0	0	0.2	0.31	0.30	0.480	0.618	failed
	CPU, LC off	2.8	0	1.8	0.36	0.28	0.456	1.028	failed
`realv7_000`	GPU, LC on	3.0	0	2.1	0.37	0.75	0.268	0.252	partial
	GPU, LC off	3.0	0	2.0	0.36	0.69	0.251	0.259	partial
	CPU, LC on	4.3	1	2.6	0.39	0.34	0.266	0.275	partial
	CPU, LC off	4.3	1	1.6	0.34	0.32	0.259	0.261	partial
`realv7_001`	GPU, LC on	2.9	0	2.0	0.42	0.71	0.451	0.477	completed
	GPU, LC off	4.7	0	1.7	0.45	0.67	1.278	0.817	completed
	CPU, LC on	2.8	0	0.8	0.40	0.33	0.575	0.438	completed
	CPU, LC off	2.9	0	2.0	0.42	0.32	0.442	0.558	completed
`realv7_002`	GPU, LC on	3.0	0	2.0	0.43	0.57	0.172	2.115	completed
	GPU, LC off	3.1	0	2.5	0.45	0.63	0.198	2.079	partial
	CPU, LC on	3.2	0	2.5	0.87	0.32	0.194	0.472	completed
	CPU, LC off	3.2	0	2.8	1.00	0.31	0.198	1.675	completed

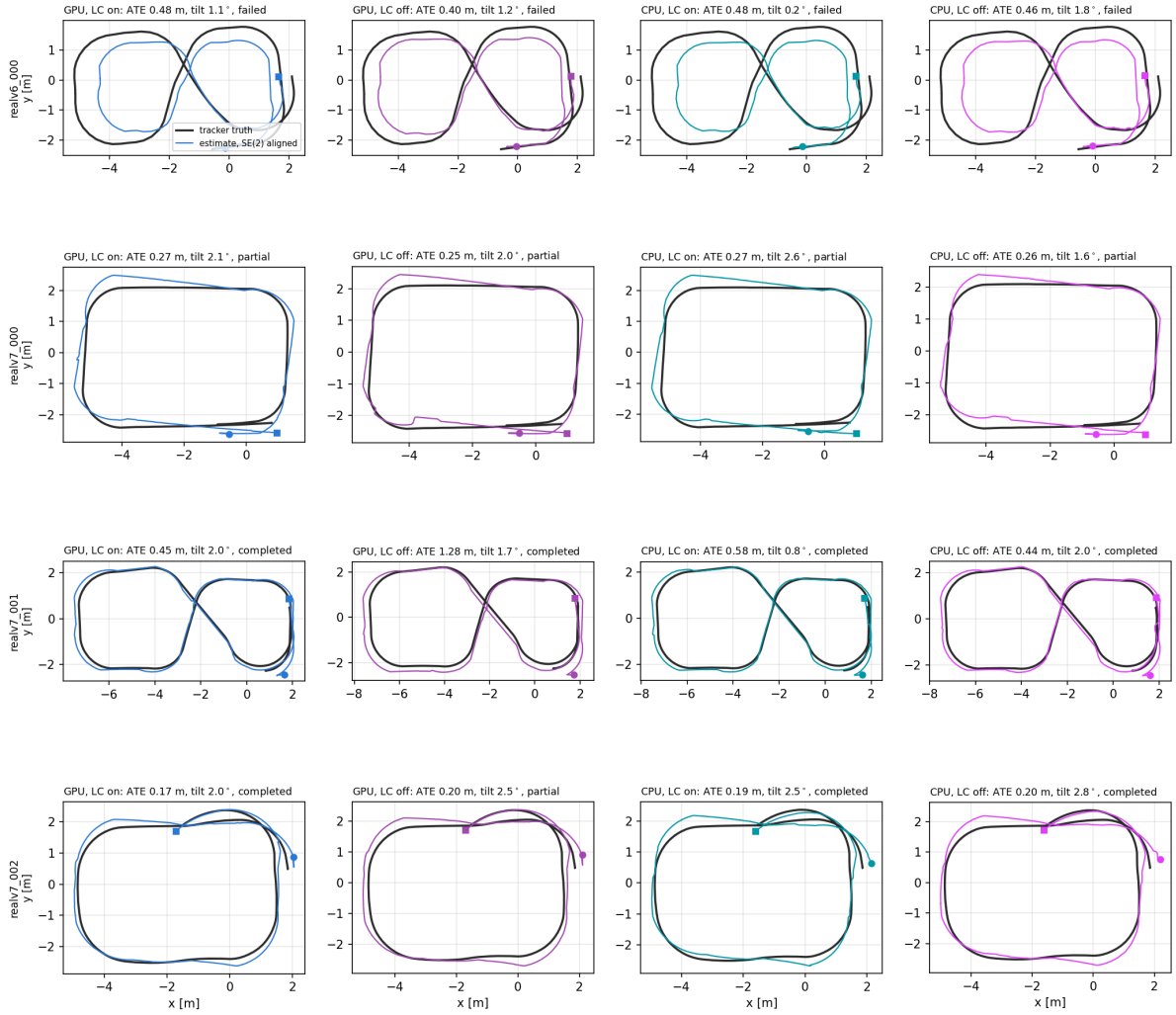


Figure 1.16: Every real-robot ORB-SLAM3 run on the four recordings that carry an independent planar reference, with fork patch 15 active, against the tracker truth after rigid SE(2) alignment. No repair of any kind is applied: these are the estimates as the node wrote them, where those of figure 1.15 had to be levelled and rotated before they could be drawn at all. Each panel gives the planar ATE, the tilt of the fitted plane and the scorer's status.

Table 1.18: VINS-Fusion on the three `dataset_realv7` recordings (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm 9.2642). Frames received against the bag's count (30 Hz over the recording), IMU use, resets, and where the estimate ended.

Recording	frames received	IMU received / used	resets	z range [m]	$\lvert p \rvert$ at end [m]
`realv7_000`	1155 of about 2663	17906 / 19053	0	12834403	37554
`realv7_001`	1541 of about 3686	24809 / 26340	0	3701	15974
`realv7_002`	1271 of about 2910	19368 / 20627	0	25937	5252